

UNIQUELY JUSTIFIABLE STRATEGIES AND MECHANISM DESIGN WITH BOUNDEDLY RATIONAL PLAYERS

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ABSTRACT. A strategy is *uniquely justifiable (UJ)* if, at each decision point, the prescribed action is the unique best response to any contingency the player may consider. We define *UJS mechanisms* as mechanisms in which the dominant strategy is also uniquely justifiable, so that boundedly rational players do not have to perform contingent reasoning. By contrast, the leading alternative, Obviously Strategy Proof (OSP) mechanisms, require players to contingently reason about their own future actions. The presence or absence of uniquely justifiable strategies neatly organizes existing evidence of adherence — or non-adherence — to (obviously) dominant strategies across mechanisms, from binary allocation problems to matching problems. We characterize the set of UJS mechanisms for binary allocation problems to show that they resemble a generalized Multiple Price List. To compare OSP and UJS mechanisms in a setting that avoids confounds, we conduct an experiment using novel elicitation mechanisms of each type. We identify a tradeoff: OSP mechanisms fail because players fail to forecast their future actions, while UJS mechanisms allow other forms of boundedly rational mistakes.

1. INTRODUCTION

Mechanism design is perhaps the most practical tool to come out of microeconomic theory, with applications ranging from auctions to assignment problems (e.g. assigning students to schools). However, mechanisms only work to the extent that people understand them, and behavioral economics has produced overwhelming evidence that people are cognitively constrained. In particular, mechanisms with dominant strategies do not work as well as theory would suggest: examples include overbidding in second price auctions (Kagel et al., 1987; Kagel and Levin, 1993), untruthful reporting in strategy-proof matching mechanisms (Chen and Sönmez, 2006), and mechanisms for the provision of public goods (Cason et al., 2006)). Furthermore, failures are observed even in single-player elicitation environments (e.g., Cason and Plott, 2014 and Danz et al., 2022). Given these limitations, a nascent

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literature has turned to developing robust mechanisms that ensure truthful revelation even with boundedly rational individuals (e.g., Li, 2017a and Pycia and Troyan, 2023).

To illustrate why it may be difficult for boundedly rational individuals to identify the dominant strategy in a mechanism, consider the ubiquitous Becker-DeGroot-Marschak (BDM) mechanism (Becker et al., 1964). In this mechanism, a player reports her willingness-to-accept (WTA) for some item she owns. She sells the item whenever a randomly chosen price exceeds her bid, receiving the randomly chosen amount in return (and otherwise keeps the item). To bid optimally, the player must reason through many contingencies. At every contingency in which the random price is above her true valuation, she can justify any bid below that price because all of these bids maximize her payoff by selling the item at the random price. Conversely, at every contingency in which the random price is below her valuation, any bid above that price is justifiable. The challenge is to think through all possible contingencies in order to realize that bidding her true valuation is optimal. In light of this complexity, it is perhaps not surprising that non-optimal bids are common with the BDM mechanism (Cason and Plott, 2014).

Li (2017a) introduces the concept of *obviously strategy-proofness* (OSP) to reduce the need for such contingent reasoning. Unlike general strategy-proof mechanisms — which require players to reason through all possible contingencies involving moves by Nature and others — OSP mechanisms allow players to compare only the best- and worst-case outcomes that can arise. The OSP mechanism for eliciting a player’s WTA takes the form of a descending clock auction: at each of a series of prices, the player chooses to either accept the price or exit the auction and keep the item. The mechanism terminates at a random price or when the player exits, whichever happens first. Such a clock auction reduces the need for contingent reasoning over moves of Nature because the the worst-case outcome from following a player’s dominant strategy is preferred to the best-case outcome from any deviation.

Critically however, OSP mechanisms, such as this clock auction and the OSP-Top-Trading Cycle matching mechanism (Troyan, 2019), do not eliminate all contingent reasoning requirements. In such dynamic mechanisms, the player must first reason contingently about her *own* future actions in order to be able to compare the best- and worst-case outcomes for each current action. For example, in the clock auction, she must recognize that she will exit when the price reaches her valuation in order to understand that she should accept the price when it is above her valuation. If she fails to think ahead, she may believe it’s possible to receive less than her valuation, which can justify exiting the mechanism early to avoid the possibility of a low payout. The potential problem here is that both choosing the price and exiting the mechanism can be justified at prices above the player’s valuation, depending on what future contingencies a boundedly rational player deems possible.

With such examples in mind, in this paper, we introduce the concept of *Uniquely Justifiable Strategy* (UJS) mechanisms which take a novel approach to eliminating the need for contingent reasoning. We first introduce the notion of a *justifiable action*, an action which can be justified as the best response to some payoff-relevant contingency — that is, to some move of Nature and sequence of a player’s own and others’ future actions for which a player’s current action affects her payoff. Importantly, the set of payoff-relevant contingencies may include non-optimal future actions of oneself or others. A *justifiable strategy* for any player then consists of some sequence of *justifiable actions*. A strategy is *uniquely justifiable* if, at each decision, it prescribes the *only* action that can be justified as a best response to *any* payoff-relevant contingency.

We define a mechanism to be a UJS mechanism if, for every player-type there exists a uniquely justifiable strategy that coincides with the dominant strategy of that player-type. To expand the scope for such mechanisms, we allow the uniquely justifiable strategy to exist either in the original game or in a simplified representation of the game in which the players treat multiple future contingencies as equivalent.¹

A UJS mechanism, when it exists, allows players to simply best-respond to any conjecture of what will happen in the future, avoiding the need to reason contingently about others’ actions (including Nature) and their own future actions. The BDM mechanism is not a UJS mechanism because many non-dominant actions (bids) are justifiable for any given contingency (randomly drawn amount). The clock auction is also not a UJS mechanism because, as discussed above, the non-dominant action of exiting early is justifiable in a contingency where the player doesn’t foresee exiting at her valuation.

In our theoretical contribution, we focus on a case of particular interest: binary allocation mechanisms that allocate a single good among players (e.g., BDM, auctions, or binary public goods) while discriminating between three or more types of any player. In this setting, the OSP mechanism takes the form of a clock auction (Li, 2017a), which, as discussed previously, requires contingent reasoning over one’s future actions. Formalizing the discussion above, we first show that in this binary allocation setting, no mechanism can simultaneously be both an OSP mechanism and a UJS mechanism. We then formally characterize the set of UJS mechanisms in this environment, showing that they take the form of a generalized multiple price list (MPL). In such a mechanism, the player makes a sequence of binary decisions between the item and a set of prices. To determine the outcome, one of these decisions is designated as ‘the choice that pays’ — either selected randomly by Nature in a standard price list, or by the actions of other players in auctions and binary public good mechanisms.

¹The simplified game allows for reasoning failures akin to cursed players in Eyster and Rabin (2005), who believe all opponent types act identically, or analogy-based expectations in Jehiel (2005), where decision nodes are grouped into “analogy classes” assumed to have identical behavior. Where we depart is in applying this logic not only to others’ decisions, but also to a player’s own future decisions.

Intuitively, the only payoff-relevant contingency for any decision is the one in which it is the ‘the choice that pays’, and hence the uniquely justifiable action is straightforward: choose the item if the price is below one’s valuation and choose the price otherwise.

The UJS concept proves valuable across multiple environments in several ways. First, it allows mechanisms that don’t require contingent reasoning about one’s future actions to exist in environments where other concepts do not. In particular, we show that the Strongly Obviously Strategy Proof (SOSP) mechanisms introduced by Pycia and Troyan (2023) cannot discriminate between three or more types of any player in any binary allocation mechanism, severely restricting their use as elicitation mechanisms and in auctions and binary public good problems. Second, UJS mechanisms are not limited to binary allocation settings – we show they exist in other settings, such as matching environments. Third, even in standard strategy-proof and OSP mechanisms, the presence or absence of uniquely justifiable strategies neatly explains existing evidence of adherence — or non-adherence — to (obviously) dominant strategies: across a range of economic environments players tend to choose dominated actions that are justifiable, but seldom choose non-justifiable actions. Although this existing empirical evidence is strongly suggestive of the importance of justifiable actions, disentangling UJS from OSP remains challenging because the better-performing mechanisms (e.g, variants of Random Serial Dictatorship) often satisfy both refinements of strategy-proofness.

To overcome this problem, we conduct an online experiment with two main goals: (i) to compare the performance of OSP and UJS mechanisms with their traditional counterparts and (ii) to identify whether or not the presence of non-dominant justifiable strategies is responsible for failures in OSP mechanisms. Binary allocation elicitation mechanisms are well-suited for these purposes. First, as noted above, OSP and UJS mechanisms are mutually exclusive in all binary allocation problems with three or more types. Second, single-player elicitation mechanisms allow us to rule out failures of strategic reasoning about others’ behavior and other-regarding preferences as sources of deviations from theoretical predictions. Finally, elicitation mechanisms are worth studying in their own right as they are crucial tools for economists, allowing us to elicit valuations, certainty equivalents (to study risk preferences), and beliefs (to study belief updating in decision-making and strategic games); as well as to study classic anomalies such as the endowment effect.

In our first three experimental treatments, we elicit subjects’ WTA for a fixed item of \$1.00 (following Cason and Plott, 2014) using the BDM mechanism, an OSP mechanism (called OSP-E), and a UJS mechanism (called UJS-E). The OSP-E and UJS-E mechanisms operate almost identically: over many periods, subjects choose between a decreasing clock-price and \$1.00. The UJS-E mechanism terminates at some randomly selected period, and the subject receives her choice in that period. The OSP-E mechanism is the same, but a subject can also

preemptively terminate the mechanism by choosing \$1.00. In both mechanisms, the weakly dominant strategy is to choose the item after the clock-price decreases to \$1.00.

Our first main empirical result is that both the UJS-E and OSP-E mechanisms greatly outperform BDM, which suggests that contingent reasoning failures play a role in the failures of the BDM mechanism. The fact that the OSP-E mechanism does better than BDM provides evidence, in addition to Li (2017a) and Breitmoser and Schweighofer-Kodritsch (2021) that OSP mechanisms can indeed outperform at least some of their non-OSP equivalents. However, we find that 60% of subjects do not play rationally in the OSP-E mechanism. Of these, more than 70% irrationally choose the dominated action (\$1.00), which is justifiable only when a subject considers a contingency which is impossible under optimal future actions (receiving less than \$1.00).

We conduct two diagnostic treatments to assess whether or not these failures of the OSP-E mechanism can be directly tied to the justifiability of the dominated action. In the first, UJ-OSP, we use a diagnostic mechanism that is identical to the OSP-E mechanism except that we modify it to make the obviously dominant action *uniquely* justifiable. We find that subjects play the dominated action significantly less often in this diagnostic treatment, which suggests that subjects justify the dominated action in OSP-E by considering future contingencies that are never reached under the dominant action. In the second diagnostic treatment, OSP-L, we provide direct evidence that subjects do so: many subjects explicitly report that they consider future contingencies that do not arise under the dominant strategy, and these subjects are more likely to take the dominated action.

Finally, we turn to the question of whether or not the UJS-E mechanism outperforms the OSP-E mechanism. We find that it does in the natural setting where exiting the OSP-E mechanism allows one to finish the experiment, but it doesn't when we force subjects to remain engaged in the task until the randomly chosen clock-price (as in the UJS-E mechanism). Thus, rather than drawing strong conclusions about which mechanism performs best, we instead suggest that our results point to a potentially unavoidable tradeoff in the design of mechanisms robust to bounded rationality. OSP mechanisms require subjects to think through their future actions. UJS mechanisms, as in an MPL, make thinking about future actions unnecessary by allowing the player to make independent choices in a price list. But, in doing so, **UJS mechanisms allow for another possible form of bounded rationality as players can switch back and forth between the price and the item. This multiple switching behavior makes up the vast majority of the failures of the UJS-E mechanism we observe.**

A large experimental literature demonstrates robust failures of contingent reasoning in a variety of settings (Charness and Levin, 2009; Carrillo and Palfrey, 2011; Esponda and Vespa, 2014; Magnani and Oprea, 2017; Martínez-Marquina et al., 2019; Ngangoué and

Weizsäcker, 2021; Esponda and Vespa, 2023).² This literature has spawned recent attempts to design mechanisms that are robust to contingent reasoning failures, specifically Li (2017a) and Pycia and Troyan (2023).³ Our theoretical contribution builds on this literature, but takes a different approach. The OSP and SOSP concepts are microfounded by a notion of simplicity in which a player doesn't need to know the structure of the game they are playing as long as they can recognize the sets of outcomes. We instead microfound simplicity by allowing a player to best-respond to any payoff-relevant contingency, rather than having to reason through all payoff-relevant contingencies.

Elicitation schemes have been tested and compared in several papers. Cason and Plott (2014) were the first to clearly demonstrate the problem with BDM mechanisms by eliciting valuations for an object of known value. Martin and Muñoz-Rodríguez (2022) replicate their finding and show that reframing BDM by providing contingency tables improves its performance. Bartling et al. (2015) use BDM to elicit valuations for a known dollar amount to study the endowment effect. A contemporaneous paper by Brown, Lui, and Tsoi (2023) also empirically compares BDM, OSP, and UJS elicitation mechanisms in a laboratory, finding that UJS mechanisms result in the highest proportion of rational behavior. In contrast to these papers, we study the reasons for failures in OSP (and more generally, strategy proof) mechanisms, and introduce the theoretical foundation for UJS mechanisms.

More generally, researchers have been interested in testing mechanisms designed to be robust to bounded rationality in a variety of economic settings (Li, 2017a; Hakimov and Kübler, 2021; Rees-Jones and Skowronek, 2018; Bó and Hakimov, 2020;2024; Hassidim et al.,2020). We revisit this empirical literature to show that the concept of unique justifiable strategies, or their absence, can explain the types of mistakes that are most commonly observed.

Ivanov et al. (2010) and Bosch-Rosa and Meissner (2020), respectively, show that subjects do not always best respond to their own actions when taken in the past or when taken simultaneously. In a contemporaneous paper, Chakraborty and Kendall (2025) provide evidence that subjects fail to perform the contingent reasoning required to forecast their own future actions correctly in simple two-period investment games. Hao and Hauser (2012) elicit probabilities using two mechanisms similar to our BDM and OSP-E mechanisms, but neither mechanism is OSP (Tsakas, 2019).

The rest of the paper is organized as follows. Section 2 introduces formal definitions, theoretical results, and hypotheses. Section 3 illustrates the applicability of the UJS concept

²Many of the failures of contingent reasoning observed in this literature can be explained by multiple actions being justifiable. For brevity, in this paper we focus on failures of mechanisms.

³See also Zhang and Levin (2017), Li (2017b), Mackenzie (2020), and Zhang and Levin (2021). For matching applications, see Troyan and Morrill (2020) and the references therein.

across a range of environments, documenting existing evidence consistent with the implications of justifiability. Section 4 describes the experimental design and Section 5 reports our empirical findings. Section 6 concludes.

2. THEORY AND HYPOTHESES

In this section, we formalize our concept of *uniquely justifiable strategies (UJS)* and compare it to the existing concepts of obviously dominant (OD) and strongly obviously dominant (SOD) strategies. For illustrative purposes, and to develop hypotheses, we first describe three binary allocation elicitation mechanisms that we utilize in the experiment: (i) BDM, (ii) OSP-E, which is based on Li (2017a)’s concept of obviously strategy-proof (OSP) implementation, and (iii) UJS-E, which is based on our concept of UJS. We then outline the potential behavioral advantages of OSP-E before discussing a potential limitation — non-dominant actions can also be justified when future actions are not optimally planned. Finally, we introduce the UJS framework and characterize the set of UJS mechanisms in the binary allocation environment.

2.1. Notation for Extensive Form Games. A finite extensive form-game, G , with consequences in X is a tuple $(N \cup \{0\}, H, \succ, \mathcal{A}, P, (\mathcal{I}_i, u_i, z_i)_{i \in N})$, where N is the set of players excluding Nature and H is a set of nodes (histories). We assume that whenever Nature (Player 0) moves, it does so prior to any player. $(\mathcal{I}_i)_{i \in N}$ partitions the set of histories into information sets for the players and z_i maps terminal histories to outcomes for player i . P is a player-function that maps non-terminal histories (and information sets) to the player who moves at that history. Suppose each player $p \in N$ has a *private type* $j \in \mathcal{M}_p$. For binary allocation mechanisms, the private type corresponds to a valuation and we will interchangeably denote it by v_j .

$\mathcal{A}$ provides the set of available actions at each information set, and u_i maps i ’s outcomes and private type to utilities. $(H, \succ)$ forms an arborescence for a partial order $\succ$ on the set of histories. Thus, we write $h_1 \succ h_2$ whenever $h_1 \in H$ succeeds $h_2 \in H$. Similarly, for $I_i, I'_i \in \mathcal{I}_i$, we say that $I_i \succ I'_i$ if there exists $h_1 \in I_i$ and $h_2 \in I'_i$ such that $h_1 \succ h_2$. Let $h_\emptyset$ be the initial history.

When making a decision at any information set, I'_i , we refer to decisions made by i at future information sets, $I_i \succ I'_i$, as being made by i ’s future self. For $a \in \mathcal{A}(I_1)$, we write $a_1 S$ to denote the modified strategy where a player plays a_1 at information set I_1 , and plays according to S at all other information sets. Let $\mathcal{S}_{-p}$ be the set of all strategies that can be played by players other than p .

We say that the weakly dominant strategies, S^i and S^j , of types i and j of Player p are *visibly different* under some strategy, S_{-p}^* , if S^i and S^j specify different actions at some

history, h_{ij, S_{-p}^*} , that is reached by both S^i and S^j under S_{-p}^* . We provide a more formal definition (7) in Appendix A.1.

Definition 1. We say that a mechanism separates all relevant types for all non-Nature players if for any player $p \in N$

- (i) There exists a set of type-contingent strategies $\mathcal{S}_p^* = \{S_p^j : j \in \mathcal{M}_p\}$ of player p such that $S_p^k \in \mathcal{S}_p^*$ is a weakly dominant strategy for type j if and only if $k = j$.
- (ii) There exists a strategy profile S_{-p}^* — comprising of one weakly dominant strategy for some type of each other player, together with Nature’s moves — such that for all type-pairs $i, j \in \mathcal{M}_p$, S_p^i and S_p^j are visibly different under S_{-p}^* .

Definition 1 captures the objective of a designer who seeks to fully extract private information from all players by separating them by type. Examples include an auctioneer who wants the allocation and payment rules to depend on the realized types, a utilitarian government deciding to build a public good when the sum of realized valuations exceeds the cost, or an experimentalist eliciting the valuation or beliefs of a subject. Part (i) requires different types to play mutually-exclusive dominant strategies. But, two different types might play strategies that only differ at an information set which is never reached under these strategies. (ii) rules out this technicality, by requiring some strategy of others (with non-Nature players playing their dominant strategy), S_{-p}^* , for which the two types are empirically distinguishable.

2.2. Examples of Binary Allocation Elicitation Mechanisms. In the experiment, we focus on single-player, binary allocation elicitation mechanisms in which each outcome either allocates a single item to Player 1 (item outcome) or does not (non-item outcome).⁴ Here, we provide examples from this setting which we will use to illustrate all theoretical concepts. However, we emphasize that all of our definitions are for arbitrary games, and our characterizations hold for all binary allocation games, which, alongside elicitation mechanisms, also encompass unit-demand auctions, unit-supply procurement auctions, and binary public-good problems.

Suppose Player 1 has a private valuation (type) $v^k \in \mathcal{M} = \{0.05l + 0.025 : l \in \mathbb{N}, 0 \leq l \leq 30\}$ for an item that she owns. Examples 1-3 describe the three binary allocation mechanisms we use in our experiment to elicit this valuation. As we explain in the following text, mechanisms 2 and 3 satisfy the OSP and UJS properties, respectively, and are named accordingly. The “-E” suffix indicates that they are elicitation mechanisms.

Mechanism 1 (BDM). *A single seller (Player 1) is asked to report a number $v \in \mathcal{M}$. Nature randomly draws a number $v^r \in \{0, 0.05, 0.10, \dots, 1.50\}$. If $v^r > v$, then the mechanism buys the item at the price v^r . Otherwise, the seller keeps the item.*

⁴Item outcomes can consist of solely the item, or some combination of the item and monetary payoffs. Non-item outcomes may consist of monetary payoffs only.

Mechanism 2 (OSP-E). *A clock-price decreases from 1.50 to 0 in increments of 0.05. At each clock-price, a single seller is asked to choose between the item and attempting to sell it at the current price. Choosing the item is the exit option: it ends the game, and the seller keeps the item. Nature ends the game at a randomly chosen clock-price v^r , and if the seller attempted to sell at that price, then the mechanism buys the item at the price v^r . This mechanism is a one-player version of a clock auction.*

Mechanism 3 (UJS-E). *Identical to OSP-E except that the game only ends when the clock-price reaches v^r (choosing the item does not end the game or let the seller exit). If the seller has chosen the item at v^r , then she keeps the item, otherwise she sells it at v^r .*

The BDM, OSP-E, and UJS-E mechanisms (1-3) separate the types, $v^k \in \mathcal{M}$ defined above. In BDM, each type v^k has a unique dominant strategy to bid v^k , whereas in OSP-E and UJS-E, each type v^k has a unique dominant strategy of choosing the item if and only if the clock-price is at or below $0.05k$.⁵

By endowing subjects with a virtual item that can be redeemed for $\$v^*$, one can induce an item-valuation of v^* and then straightforwardly measure the performance of each type-separating mechanism as the percentage of subjects who choose the dominant strategy for their induced type, v^* (Cason and Plott, 2014). Doing so leads to our first testable (null) hypothesis:

Hypothesis 0 (Equal). *Subjects with an induced valuation are equally likely to play their respective dominant strategies in Mechanisms [1]-[3].*

Despite the apparent similarity of the mechanisms (especially OSP-E and UJS-E), in the following sections we argue that these mechanisms might perform very differently for boundedly rational players.

2.3. Obviously Strategy-Proof (OSP) Elicitation Mechanisms and a Potential Limitation.

Following Li (2017a), a strategy S_i is *obviously dominant* (OD) if for any deviating strategy S'_i , starting from the earliest information set where S_i and S'_i diverge, the worst possible payoff from S_i is at least as good as the best possible payoff from S'_i , where the best/worst case outcomes are determined over all possible future actions of other players including Nature (S_{-i}), and i 's future actions are dictated by S_i . OD is a notion of robustness to opponents' play.

In OSP-E (mechanism 2), any type v^k has a OD strategy, S^k , to choose to stay in for all clock-prices above v^k and to choose the item at all prices lower than v^k . For example, fix a clock-price $q > v^k$: deviating from S^k by exiting at q guarantees v^k whereas the minimum

⁵For OSP-E and UJS-E, we require $S_{-1}^* \in \{0, 0.05\}$ to meet condition (ii), but we note that the distribution over Nature's moves can be chosen such that these states are reached with arbitrarily high probability.

payoff from continuing with S^k guarantees at least v^k (when $v^r \leq v^k$) and potentially more (when $v^r > v^k$). Thus, OSP-E is obviously strategy-proof (OSP): an obviously dominant strategy exists for every type of player, and the strategies are visibly different.

S^k also describes the dominant strategy in UJS-E, but it is no longer OD for types with $v^k < 1.5$. Fix a clock-price $q > v^k + 0.05$: the minimum payoff from S^k is v^k (when $v^r < v^k$). Now consider the deviation strategy that coincides with S^k everywhere except at the current clock-price q , where the player chooses the item instead. This deviation yields a maximum payoff of $q - 0.05 > v^k$ (for $v^r = q - 0.05$), thus failing the OD requirement. Similarly, players in BDM (Example 1) do not have an OD strategy.

Consider a boundedly rational player who recognizes the set of outcomes that are feasible under strategies S^1 and S^2 , but cannot perform the contingent reasoning required to understand how, *state by state*, the outcomes after playing S^1 are related to the outcomes after playing S^2 .⁶ Such a player would only be able to play the dominant strategy in OSP-E, even though the mechanisms 1-3 have equivalent dominant strategies. The possibility of such boundedly rational behavior gives us the following alternative to Hypothesis 0:

Hypothesis 1 (OSP). *Subjects are more likely to play their dominant strategies in OSP-E than in BDM (Hypothesis 1.i) and UJS-E (Hypothesis 1.ii).*

Despite being robust to one’s opponents’ actions, OSP has a potential limitation: rational play requires each player to correctly forecast *their complete optimal contingent plan of actions (strategy)*. Even if a strategy is obviously dominant, players who struggle with contingent reasoning may not be able to reason through actions they take far in the future, and therefore fail to choose optimally in the present. Current actions that are dominated may appear optimal – or justifiable, as we define later – when evaluated under suboptimal future play.

Example 1. In OSP-E, if a player with valuation $v^k = 1$ believes that she could exit at a price below 1 and thus receive a payoff less than her valuation, then she may find it optimal to exit the mechanism before the clock-price reaches 1.

This simple observation about OSP-E summarizes our central behavioral hypothesis: even in games with an OD strategy, boundedly rational players might think a dominated action is optimal, because they fail to forecast their optimal path of future play. This potential failure in contingent reasoning about one’s own future play is consistent with the recent experimental literature (see Chakraborty and Kendall, 2025, and the references within).

2.4. Strongly Obviously Strategy-Proof (SOSP) Mechanisms can be Restrictive. Pycia and Troyan’s (2023) *strongly obviously dominant (SOD)* concept strengthens OD further,

⁶See Li (2017a) for a more technical explanation of the bounded rationality that OSP games allow for.

by requiring robustness to *any* future contingencies one can consider. Under SOD, the best and worst case outcomes are determined over all possible future actions of other players, including Nature (S_{-i}) and a player's own future actions. Mechanisms are *strongly obviously strategy-proof* (SOSP) when a strongly obviously dominant strategy exists for each type.

Unfortunately, as we show in Proposition 1, no binary allocation mechanism — elicitation or otherwise — can satisfy SOSP and also separate $n > 2$ types. For an SOSP mechanism to separate all relevant types, each strategy, S^k , in Definition 1 must be SOD (not just weakly dominant). Consider a general mechanism that can assign $q \in \{0, 1, 2, \dots, m\}$ units of the item to any participant — the binary allocation mechanism being the special case of $q = 1$. We show that for an SOSP mechanism to separate n types, q has to be at least as large as $n - 1$.

Proposition 1. *i) When $m < n - 1$, there exists no SOSP mechanism that transacts up to m units of the item and separates all n types of player p .*

ii) When $m = n - 1$, a SOSP mechanism exists that separates all n types of player p using up to m units of the item. In particular, let (k, m) denote the outcome with k items and $\$m$. The mechanism that offers n actions $(b_i)_{i=1}^n$ such that b_n pays $((n - 1), 0)$ and b_i pays $\left((i - 1), \sum_{j=i}^{n-1} \frac{v_{j+1} + v_j}{2}\right)$ for $i \in \{1, \dots, n - 1\}$ is SOSP and separates all n relevant types of a player p .

Proposition 1 shows that, for practical applications in which up to one item is to be allocated ($m = 1$), SOSP elicitation mechanisms do not exist for $n > 2$ types. The proof leverages Pycia and Troyan's (2023) result that an SOSP mechanism must offer each player a single choice from a menu of options. We then show that separating types (definition 1) requires that a higher-valuation type's best option from the menu must contain strictly more items than a lower type's best option, which implies no more than two types can be separated in an SOSP binary allocation mechanism.

2.5. Uniquely Justifiable Strategies (UJS). Identifying a contingent plan of actions is cognitively demanding — let alone finding the plan that is optimal among all such plans. This complexity motivates our new concept: *Uniquely Justifiable Strategies (UJS)*. Rather than expecting players to identify the optimal plan, a uniquely justifiable strategy ensures that simply best responding to *any* future plan of actions is sufficient to arrive at the dominant strategy for their type.

We define a UJS mechanism in two steps. First, to allow best responding to any future plan of actions, we introduce the notions of *justifiable action* and *justifiable strategy* (Definitions 3, 4), which are more permissive concepts than dominant strategies. Next, to accommodate individuals who might fail to do contingent reasoning, and hence adopt a simplified representation of the original game, we introduce the concept of a *simplified game* (Definition 5). Finally, we define a mechanism to be a *UJS (Uniquely Justifiable Strategy)* mechanism

(Definition 6) if each player's dominant strategy of the original game is also her uniquely justifiable strategy in some simplified game.

2.5.1. Formal Definitions of Justifiable strategies and Uniquely Justifiable Strategy (UJS) Mechanisms. We begin by defining a *contingency* at $I_p \in \mathcal{I}_p$ for player p (or Pp) as a joint conjecture (h, S_p, S_{-p}) about the history ($h \in I_p$), others' actions (S_{-p}) and one's own future actions (S_p). We consider a boundedly rational player who forms a conjecture about the actions taken at all future information sets, including the ones where she will move herself – as if treating her future self as a different player. Importantly, when making the current choice, such a player might consider future contingencies where she acts irrationally, because she is unable to predict the rational course of actions in complex sequential choice environments.

Let $\mathcal{O}_p^G(h, S_p, S_{-p}) = \cup_{a \in \mathcal{A}^G(h)} z_p^G(h, a, S_p, S_{-p})$ be the set of all possible outcomes for player p in game G under contingency (h, S_p, S_{-p}) . We say that an outcome is *undominated* at a contingency, if there does not exist an outcome reachable under that contingency that is preferred by all types of that player. In binary allocation games, this would imply an outcome that either offers an extra item or strictly higher money than all other outcomes.

To allow for behavior that best responds to arbitrary contingencies at every step, we introduce the notions of *justifiable action* and *justifiable strategy*.

Definition 2. (h, S_p, S_{-p}) is a *payoff-relevant contingency* at $I_p \in \mathcal{I}_p$ for Player p if $h \in I_p$ and $|\mathcal{O}_p^G(h, S_p, S_{-p})| > 1$. If there is at least one payoff-relevant contingency at an information set $I_p \in \mathcal{I}_p$, we call it a *payoff-relevant information set*.

Definition 3. Action $a^* \in \mathcal{A}(I_p)$ is *justifiable* for type k of Player p at $I_p \in \mathcal{I}_p$ if

$$a^* \in \cup_{(h, S_p, S_{-p})} \{ \arg \max_{a \in \mathcal{A}(I_p)} u_p(h, a, S_p, S_{-p}, k) \}$$

i.e, the action is utility-maximizing for some contingency (h, S_p, S_{-p}) at I_p . Action $a^* \in \mathcal{A}(I_p)$ is *uniquely justifiable* for type k of Player p at I_p if

$$\{a^*\} = \cup_{\{(h, S_p, S_{-p}): h \in I_p, |\mathcal{O}_p^G(h, S_p, S_{-p})| > 1\}} \{ \arg \max_{a \in \mathcal{A}(I_p)} u_p(h, a, S_p, S_{-p}, k) \}$$

i.e, it is the only *justifiable* action at any payoff-relevant contingency considered at I_p .

Definition 4. S_p^* is a *uniquely justifiable strategy* for type k of Player p if every $S_p^*(I_p)$ is a *uniquely justifiable action* for type k of Player p at every payoff-relevant information set, I_p .

The notion of a justifiable strategy captures a form of myopic bounded rationality: at each step, the player best responds to some contingency under various hypothetical scenarios, without evaluating whether those contingencies — in particular, her future choices — are themselves consistent across steps. Doing so allows actions to be justified by beliefs about

future behavior that may in fact never materialize. A justifiable strategy then describes the complete course of play followed by such a player.

Although the set of justifiable actions at a given information set may be large — reflecting the permissiveness of this concept — our goal is to construct mechanisms where this multiplicity collapses for each player to some uniquely justifiable strategy. Crucially, this collapse is allowed to occur either in the original game, or in some *simplified* version of the game that reflects a player’s limited ability to engage in contingent reasoning.

Definition 5. We call G' a Pp -simplified game of G , if G' replaces Player p ’s information partition by a coarser information partition.⁷ For $j \neq p$, $\mathcal{I}_j^{G'} = \mathcal{I}_j^G$, $\mathcal{S}_j^{G'} = \mathcal{S}_j^G$.

The set of possible Pp -simplified games of G includes G itself, as well as any game which arises by pooling together one or more of Player p ’s information sets. A strictly coarser partition reflects Player p ’s simplified view of the original game, where she treats a wider range of contingencies as equivalent and plans to act identically across them. Doing so captures a failure of contingent reasoning — akin to cursed players in Eyster and Rabin (2005), who believe all opponent-types in a Bayesian game act identically, or to players in Jehiel (2005), who group others’ decision nodes into ‘analogy classes’ and believe that behavior is identical across all nodes within each class. Where we depart is in applying this logic not only to others’ decisions, but also to a player’s own future decisions. We can now formally define a UJS mechanism.

Definition 6. $G = (N, H, \succ, \mathcal{A}, P, (\mathcal{I}_i, u_i, z_i)_{i \in N})$ is a UJS mechanism if for every player $p \in N$

- (i) There exists a set of $|\mathcal{M}_p|$ type-contingent strategies $\mathcal{S}_p^* = \{S_p^j : j \in \mathcal{M}_p\}$ such that $S_p^k \in \mathcal{S}_p^*$ is a weakly dominant strategy for type j if and only if $k = j$.
- (ii) There exists S_{-p}^* — comprising one weakly dominant strategy for some type of each other player, together with Nature’s move — such that for all pairs $i, j \in \mathcal{M}_p$, S_p^i and S_p^j are visibly different under S_{-p}^* .
- iii) There exists a Pp -simplified game $G' = (N, H, \succ, \mathcal{A}^{G'}, P, (\mathcal{I}_i^{G'}, u_i, z_i)_{i \in N})$ of G , such that the set of strategies $\{S'^j : j \in \mathcal{M}\}$ in G' are uniquely justifiable for types $j \in \mathcal{M}$. Further, for any $S_{-p} \in \mathcal{S}_{-p}$, strategy S'^j leads to the same terminal histories in G' as S^j from (i) does in G .

A UJS mechanism retains the properties (i)-(ii) of Definition 1 that allow types to be separated. Condition (iii) ensures that players do not have to perform contingent reasoning: they are steered — through the uniquely justifiable strategy — toward the dominant strategy of the original game.

⁷Partition P_1 is finer than P_2 (or equivalently, P_2 is coarser than P_1) if every set $S \in P_1$ is a subset of a set $T \in P_2$ (which allows $P_1 = P_2$).

FIGURE 2.1. UJS-E versus Simplified UJS-E

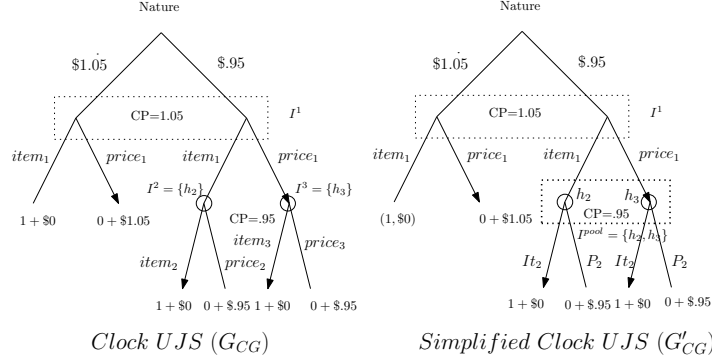


FIGURE 2.2. Contingencies (cols) and actions (rows) at $CP = 1.05$ in G'_{CG}

	$1.05It_2$	$1.05P_2$	$.95It_2$	$.95P_2$
$item_1$	1.025	1.025	1.025	.95
$price_1$	1.05	1.05	1.025	.95

Example 2. Consider the truncated UJS-E game G_{CG} (see Figure 2.1) which has only two clock-prices (CP henceforth), \$1.05 and \$0.95. P1's outcomes are written as an indicator for receiving the item (1 or 0) + a dollar amount. We show that this game is UJS by constructing a corresponding P1-simplified game G'_{CG} , where all decision nodes at a given clock price are pooled into a single information set: P1 treats all histories at the same price as indistinguishable and expects to behave identically at all of them.

Suppose the player values the item at $v^* = 1.025$. In Table 2.2, we present the four contingencies at $CP = 1.05$ in G'_{CG} . Each contingency jointly describes the history (Nature's previous move of 1.05 or .95) and one's own action (It_2 or P_2) at I^{pool} . Only the first two contingencies are payoff-relevant and $price_1$ is the uniquely justifiable action at I_1 . When the player reaches I^{pool} , she again has a uniquely justifiable action, It_2 . As a result, the player forms a uniquely justifiable strategy in G'_{CG} , which reaches the same terminal histories as the dominant strategy in the original game G_{CG} .

UJS-E (3) is a UJS mechanism. It admits a simplified game like G'_{CG} , where all non-terminal histories at the same clock price are pooled into the same information set. In this simplified representation, best responding to any payoff-relevant contingency leads the player to choose the dominant strategy of the original game.

By contrast, OSP-E cannot be a UJS mechanism. Intuitively, this is because the exit action is also justifiable at a price above the player's valuation under any contingency in which the player does not foresee exiting at her valuation. The following proposition establishes this mutual exclusivity formally for all binary allocation mechanisms.

Proposition 2. *If the binary allocation mechanism, G , is a UJS mechanism separating $n \geq 3$ types, then it cannot be a personal clock auction, and if the game G is a personal clock auction, it cannot be a UJS mechanism separating $n \geq 3$ types.*

The BDM mechanism is also not a UJS mechanism. When G is a BDM (1), the only possible simplified game G' is identical to G , since Player 1 in G already has the coarsest possible information partition. But, G offers no simplified representation in which the dominant strategy becomes uniquely justifiable. For example, suppose the player has valuation, $v^* = 1$. If she considers a contingency with $v_r \geq 1.05$, bidding any $v < v_r$ is justifiable. Thus, the player faces a large set of justifiable strategies and must reason through all possible contingencies to discover that bidding v^* is dominant.

This leads to our next hypothesis.

Hypothesis 2 (UJS). *UJS-E outperforms BDM (Hypothesis 2.i). UJS-E performs better (worse) than OSP-E if uniquely justifiable strategies capture the relevant aspects of bounded rationality more (less) effectively than obvious dominance (Hypothesis 2.ii).*

Although the OSP-E mechanism is not a UJS mechanism, a diagnostic variant exists that makes choosing the price uniquely justifiable in the original game – that is, without any simplification. We leverage this diagnostic variant in the experiment to identify whether or not the lack of a uniquely justifiable strategy is responsible for any observed failures of the OSP-E mechanism.

Specifically, consider a variant of Example 1 in which the player is forced to choose the item at all clock-prices below her valuation:

Mechanism 4 (UJ-OSP-D). *Identical to OSP-E (2), but the seller must exit at the clock-price of 1.*

At any clock price, $q > 1$, continuing is the uniquely justifiable action for players with $v^* = 1$. Suppose such a player considers a contingency where Nature’s action is q^* . At any contingency with $q^* \leq 1$, current actions are payoff irrelevant by design, so here onwards we focus on $q^* > 1$. If $q^* = q$, then continuing is the uniquely justifiable action at q . If $q^* < q$ and if the player’s future strategy is to continue until q^* , then continuing is again uniquely justifiable at q . If $q^* < q$ and if the player’s future strategy exits at or before q^* , then either action at q leads to the same outcome, so that the contingency is payoff-irrelevant. Thus, continuing is the uniquely justifiable action at all prices $q > 1$.

Unfortunately, this variation cannot be used to separate the valuations of all types: it does not make choosing the price uniquely justifiable for types $v^* > 1$, and forces all types $v^* < 1$ to play identically. This mechanism is therefore not practically useful for elicitation, but is diagnostic. Comparing its performance to that of OSP-E forms our final hypothesis:

Hypothesis 3 (UJ-OSP-D>OSP-E). *Subjects with $v^* = 1$ are less likely to exit at clock-prices above 1 in UJ-OSP-D than in OSP-E if the lack of a uniquely justifiable strategy causes failures in the OSP-E mechanism.*

2.5.2. *General Characterization of UJS Mechanisms in Binary Allocation Environments.* Here, we show that UJS mechanisms in binary allocation environments take a generalized form of a price list. A price list offers a simple structure: at each decision point, the player chooses between two actions—one that delivers the item, and another that forgoes the item in favor of a fixed monetary payoff. Crucially, each action is dedicated to a specific kind of outcome — either the item or money. A standard price list, which is slightly different than the UJS-E mechanism because Nature’s action is revealed only after the player has made binary decisions at all clock prices, is also a UJS mechanism (see example 11 in Appendix C). Not only are such price lists examples of UJS mechanisms, but the converse also holds in a precise sense. The next result shows that by simply removing redundant actions that are not justifiable for any type, any binary allocation UJS mechanism and its associated simplified game G' can be reduced essentially to a price list: a sequence of binary choices between an item outcome and a monetary outcome.

For any game G and a simplified game G' that makes it a UJS mechanism, the result is shown in two parts. The first part establishes that for a UJS mechanism, at any information set in G' , any action must always deliver the same kind of outcome — either item or non-item — regardless of the contingency, i.e, how others (including Nature) move or what future actions the player considers. This allows the higher types to uniquely justify the action leading to the best available item outcome and lower types to uniquely justify the action leading to the best available non-item outcome. The second part uses this insight to prune all but two actions from G' that don’t yield the best item or non-item outcome, yielding a pruned version of the simplified game — denoted $(G^*)'$. Finally, from the pruned simplified game $(G^*)'$, we retrieve a pruned original game G^* , which itself is a UJS mechanism and resembles a price list.

Proposition 3. *Suppose subjects have quasilinear preferences. Suppose the binary-allocation mechanism G is a UJS mechanism. Take Player p with $n \geq 3$ types. Then, the following must hold:*

(i) *Exclusivity of Outcome Type for Actions: In the corresponding Pp -simplified game G' , fix any information set I_p at which not all Player p -types have identical uniquely justifiable actions. Then if any action $a \in \mathcal{A}(I_p)$ yields a undominated item (or non-item) outcome under some payoff-relevant contingency, then it must yield undominated item outcomes (or non-item outcomes respectively) under all payoff-relevant contingencies..*

(ii) *Pruned UJS Mechanism with Two Actions per Information Set: Assume that if a payoff-irrelevant information set exists for player p in G , then the dominant strategies for all types,*

$S_p^k \in \mathcal{S}_p^*$, prescribe the same action there. One can prune Player p 's set of actions (and information sets reached by those actions) in G to arrive at a pruned game G^* (along with its corresponding simplified game $(G^*)'$), such that G^* is also a UJS mechanism and contains at most two actions per information set for Player p . Whenever two actions are present at an information set for Player p in $(G^*)'$, one leads exclusively to item outcomes and another to non-item outcomes.

This characterization applies to all binary allocation games, as we demonstrate below with UJS mechanisms for auctions (3) and binary public goods (4). Part (i) of the result provides guidance on the structure of the Pp -simplified game, G' : the information sets should be coarsened until a player's actions lead exclusively to either the item outcome or the non-item outcome. Part (ii), when applied to each player, one by one, creates a pruned UJS mechanism where each player has at most two actions per information set.

3. EXISTING EVIDENCE FOR THE ROLE OF UNIQUELY JUSTIFIABLE STRATEGIES ACROSS ENVIRONMENTS

In this section, we illustrate the applicability of UJS through a progression of well-known mechanisms — starting with private-value auctions, moving to binary public-good provision, and finally to matching in school choice models. In each case, we show how the UJS property helps to explain why some mechanisms perform better than others in practice, and how it can guide the design of novel alternatives.

Example 3. (UJS auction) A Second-Price Auction, like BDM, is known to generate substantial deviations from the dominant strategy of truthful bidding (Li, 2017a). One possible explanation is that this auction does not satisfy the UJS property: in contingencies where the highest bid among others is below a player's valuation, any bid at or above her valuation yields the maximal payoff, so truthful bidding is not the uniquely justifiable action. If this lack of uniqueness contributes to overbidding, the following UJS mechanism could improve performance.

Each player, with an integer private valuation between 1 and M , completes an individual price list over an ordered set of prices $\mathcal{Q} = \{0.5, 1.5, \dots, M + .5\}$. At every price $q \in \mathcal{Q}$ she answers Yes/No to “Would you buy at price q ?”. Let $q_i^* \in \mathcal{Q}$ denote the maximum q at which player i answers Yes. The good is allocated with equal probability among all the bidders i who say Yes at the price, $\max_{j \neq i} q_j^*$.⁸ Conditional on receiving the good, player i pays $\max_{j \neq i} q_j^*$. Players pay zero otherwise. Intuitively, each player completes a price list where the ‘row that pays’ is not determined not by Nature, but by other players’ actions.

This mechanism is a UJS mechanism: for any player i , consider the Pi -simplified game where

⁸If choices are non-monotonic because of multiple-switching, the item might remain unallocated.

all of i 's decisions at any fixed price q belong to the same information set. At price q , all payoff-relevant contingencies (others' actions) must satisfy $\max_{j \neq i} q_j^* = q$, and at any such contingency, the uniquely justifiable action is Yes if $v_i > q$ and No if $v_i < q$, thus leading to her dominant strategy.⁹ Like the second-price auction, this UJS mechanism implements the utilitarian-efficient allocation rule — assigning the item to the highest-valuation bidder — in dominant strategies, with the winner paying the second-highest valuation. When bidders have $n \geq 3$ possible valuation types, no SOSP auction can guarantee the same (Proposition 1).

Example 4. (UJS mechanism alternative for Vickrey-Clarke-Groves (VCG) in binary public goods) Similarly, the VCG mechanism for public project provision has truthful reporting as a dominant strategy, but it fails to have a UJS. To fix ideas, consider a setting with n players deciding whether to build a public project costing C . Each player's type v_i is their private valuation for the project, and is an integer between 1 and M . Under the VCG mechanism, the project is built if players' reported values, a_i , add up to at least C , and each player pays $q_i = \max\{0, C - \sum_{j \neq i} a_j\}$ in this case. In any contingency where the project is built when i announces truthfully, it is also built (and i receives the same net utility) when i overstates their value. Thus, truthful reporting is not i 's uniquely justifiable action.

An alternative to VCG which is a UJS mechanism is as follows. Each player i completes an ordered individual price list over a grid $\mathcal{Q} = \{0, 0.5, 1.5, \dots, M - .5\}$: for every $q \in \mathcal{Q}$ she answers *Yes/No* to “*Would you fund the project at price q ?*”. Let $q_i^* \in \mathcal{Q}$ denote the maximum q at which i answered *Yes*. After all lists are submitted, the designer computes, for each player i , $\hat{q}_i = \max\{0, C - \sum_{j \neq i} q_j^*\}$. The project is funded if and only if each player i answered Yes at price $\hat{q}_i$; if funded, player i pays $\hat{q}_i$. Otherwise, the project is not funded and no one pays. This mechanism is a UJS mechanism. For any participating player, at q , payoff-relevance requires $q = \hat{q}_i$, and the uniquely justifiable action is Yes if and only if $q < v_i$ — which aligns with the player's dominant strategy in the original mechanism. Like VCG, this mechanism separates all types and implements the utilitarian efficient rule, “*Build the project if and only if $\sum_i v_i \geq C$* ”, in dominant strategies, whereas there is no SOSP mechanism which guarantees the same (as implied by Proposition 1, provided the type space is non-trivial - see Appendix C).

Allowing participants to play justifiable rather than dominant strategies in VCG mechanisms also provides a novel explanation for prior experimental findings. Cason et al. (2006) compare the performance of the VCG mechanism for binary versus continuous public goods. In their first treatment with binary public goods, nearly all 25 reports are justifiable, and only about 50% of outcomes coincided with the dominant-strategy equilibrium, even after repeated play. By contrast, in their second treatment — where the public good is continuous and induced

⁹This mechanism is not OSP.

preferences are single-peaked — the set of justifiable actions fell to just 3 out of 25, and 81% of outcomes coincided with the dominant-strategy equilibrium. This pattern is precisely what our framework predicts: the fewer justifiable actions, the closer observed play aligns with dominant-strategy outcomes.

Example 5. (Deferred Acceptance (DA) and Top Trading Cycles (TTC) are not UJS mechanisms): Consider a standard one-to-one matching market with three students $S = \{s_1, s_2, s_3\}$ and three schools A, B, C , each with capacity 1. s_2 prefers $A \succ B \succ C$. Table 1 describes s_2 ’s DA (and TTC) outcomes for three different reports under fixed school priorities, π , and a given payoff-relevant contingency.¹⁰ DA (and TTC¹¹) are not UJS mechanisms for s_2 in this example. First note that $G' = G$ as there is no coarser partition. Still, truthful reporting is not uniquely justifiable for s_2 as multiple reports yield her top choice B .

School Priority	Contingency (s_1, s_3 reports)	s_2 ’s reports	s_2 ’s Outcome
$\begin{cases} \pi_A : s_1 \succ s_2 \succ s_3 \\ \pi_B : s_1 \succ s_3 \succ s_2 \\ \pi_C : s_1 \succ s_2 \succ s_3 \end{cases}$	$\begin{cases} s_1 : A \succ C \succ B \\ s_3 : C \succ A \succ B \end{cases}$	$B \succ A \succ C$	B
		$A \succ B \succ C$	B
		$C \succ B \succ A$	C

TABLE 1. We describe s_2 ’s DA (and TTC) outcomes for three different reports under fixed school priorities, π , and a payoff-relevant contingency. s_2 does not have a uniquely justifiable report, since multiple reports yield s_2 ’s top choice, B .

Based on our model of justifiable choices, we can explain why s_2 might switch her top two choices — the modal manipulation observed in multiple experimental studies (e.g., Hakimov and Kübler, 2021). Misreporting the top school becomes justifiable for s_2 only because she is not top priority. By contrast, suppose s_1 preferred A over all other schools: given she holds top priority at every school, every justifiable report by s_1 would necessarily rank A first. Thus, if decisions are indeed guided by justifiable actions, participants with unanimously high priorities should be much more likely to report their top option truthfully. This is consistent with the experimental findings of Rees-Jones and Skowronek (2018) and Bó and Hakimov (2020), who show that higher average priority strongly predicts truthful reporting, and with the field evidence from Hassidim et al. (2020) that shows Israeli applicants with lower priority (due to weaker grades) are more likely to submit dominated rank-order lists under DA than applicants with higher priority.

Example 6. (Explaining OSP-TTC deviations) The OSP-TTC mechanism (Trojan, 2019) works by tentatively assigning each object to its highest-priority participant (“owners”). Each owner is asked, one assigned object at a time, whether that is her favorite object. If she says *Yes* to any object, she receives it. If she says *No* to all assigned objects, the mechanism

¹⁰Unlike preferences, priorities are non-strategic and determined in advance.

¹¹See Appendix C.

moves on to the next owner. If all owners say *No* to all, then each owner is asked to point to one object among the objects she does not own. With each object also pointing to its owner, the mechanism looks for cycles. Participants in a cycle receive the object they pointed to and exit. Priorities are then updated based on the exits, and the process repeats until all objects are assigned.

Now compare two cases. If an owner is the top priority at her favorite remaining object, answering *Yes* is both the uniquely justifiable action and part of the obviously dominant strategy. If instead she is not the top priority at her favorite remaining object, then both *Yes* and *No* are justifiable (so this mechanism is not a UJS mechanism), but only *No* is part of the obviously dominant strategy. In experiments by Bó and Hakimov (2024), deviations from the OD strategy track justifiable actions: in case 1, where OD aligns with justifiable actions, subjects play the OD action (*Yes*) 93% of the time, but in case 2 where it does not, subjects do so (*No*) only 56% of the time.

Example 7. (SOSP Serial Dictatorship is a UJS mechanism): Suppose there is a finite set of players (say students), N , and a finite set of items (say schools), H , with $|H| = |N|$. Each player $p \in N$ holds strict preferences $\succ_p$ over H . The SD mechanism visits players in some fixed order π ; when player p is called, she is allocated the item she chooses among those remaining. Truthful reporting (i.e., picking the top available school at one's turn) is not only the (strongly obviously) dominant strategy, but also, the uniquely justifiable action in the Pp-simplified game $G' = G$.

Li (2017a) compares two variants of random serial dictatorship (RSD) in an experiment with common-value prizes. *OSP-RSD*, like Example 7, has each player choose a prize directly in turn, after observing which prizes remain¹²; it is a UJS mechanism. SP-RSD instead used the strategy method: players submit a *full ranked list* after learning their priority score, with allocation then proceeding in priority order. Although truthful ranking by value is a dominant strategy in both mechanisms, SP-RSD is not a UJS mechanism: for any player without top priority, there are contingencies in which the highest-value prize is already taken, so ranking the second-best prize first is justifiable, and truth-telling is no longer uniquely justifiable. The concept of UJS would thus predict more truthful reporting in OSP-RSD than in SP-RSD, which is what Li (2017a) finds. Furthermore, the most common misreport in SP-RSD rankings is swapping the top two prizes, which is exactly the kind of deviation that is justifiable for subjects with lower priority.

As we've seen, some mechanisms are both UJS and OSP (like OSP-RSD), some are neither (TTC, DA), and in some environments OSP mechanisms appear not to exist for $n \geq 3$ types

¹²To ensure ex-ante fairness, RSD first draws a random priority order π and then applies the induced SD. In Li (2017a), subjects were informed of the realized π before play began, making it operationally equivalent to SD with known priority.

(e.g., efficient binary public good provision). To compare the relevance of the UJS and OSP properties in isolation, we need an environment where UJS and OSP mechanisms exist, can be framed similarly for comparability, and are mutually exclusive. Binary allocation mechanisms provide mutual exclusivity (as shown in Proposition 3). Within this class, the single-player elicitation mechanisms OSP-E and UJS-E (Examples 2 and 3) allow us to further rule out failures of strategic reasoning about others' behavior and other-regarding preferences as sources of deviations from theoretical predictions. We turn to their empirical performance in the next section.

4. EXPERIMENTAL DESIGN

Our experiment has two main goals. First, we aim to compare the performance of new elicitation mechanisms (OSP-E and UJS-E) to the standard BDM mechanism, and to each other (Hypotheses 0, 1, 2). Second, we seek to evaluate the importance of multiple justifiable strategies in OSP mechanisms (Hypothesis 3).

The experimental design allows us to interpret our results in light of our theory - hypotheses based on potential forms of boundedly rational behavior. Where possible, we kept the implementation details constant across treatments, but readily acknowledge that comparisons of BDM and the other mechanisms, in particular, jointly test differences in implementation details along with theoretical differences.

4.1. Main Treatments. To achieve the first goal, we designed three main treatments, each consisting of a single task using Mechanisms 1-3. As such, all comparisons of mechanisms are *between* subjects. In each treatment, we induce a valuation of \$1 by endowing subjects with a virtual item that can be redeemed for \$1.00 or sold to the experimentalist. Using an item of known value allows us to straightforwardly test the performance of each mechanism, as in Cason and Plott (2014).

The first treatment, BDM, uses the BDM mechanism and follows a standard implementation. Similar to Example 1, subjects were told to choose the value between \$0 and \$1.50 (in a multiple of \$0.01) at which they are willing to sell their \$1.00 item. The computer then randomly and uniformly drew a value in this interval. The subject kept the \$1.00 item if the random value was less than their WTA, and otherwise received the random value. The instructions were a computerized version of those in Plott and Zeiler (2005). Importantly, as in their instructions, subjects were *explicitly* told that they should bid their valuation and were provided with several examples explaining why doing so is optimal.

The second treatment, OSP uses the OSP-E mechanism (2), which directly implements Li's (2017a) concept of obviously strategy-proof mechanisms in an individual decision-making setting. OSP-E utilizes a clock-price which begins at \$1.50 and decreases by \$0.05 each period until reaching \$0. In each period, a subject can either

(i) ‘Try to Sell’: attempt to sell their \$1.00 item at the given price, knowing that the computer has previously (uniformly) randomly chosen the price at which it will purchase the item, or,

(ii) ‘Exit the Market’: permanently exit and choose to keep the item as the final outcome.

The clock-price decreases until it reaches the randomly chosen round where the computer buys the item, or until the subject chooses to exit, whichever is earlier. We chose a starting price of \$1.50 to reduce the amount of censoring that occurs in the mechanism; censoring occurs when the randomly chosen clock-price stops the mechanism prior to the subject ever having chosen the item.¹³

The third treatment, UJS, uses the UJS-E mechanism of Example 3.¹⁴ It uses the same set of clock-prices as the OSP-E mechanism. In each period, subjects choose between ‘Try to Sell’ and ‘Do not Sell’. The task only ends when the randomly chosen price is reached.

In both the UJS and OSP treatments, after making a choice, we reminded subjects of the potential payoff consequences of their choice.¹⁵ They could then either confirm their choice, or revise it. Asking for confirmation ensures choices are not accidental clicks or trembles. Unlike in the BDM treatment, in neither the OSP nor the UJS treatments were subjects informed of the optimal strategy: choosing the price at all clock-prices above \$1.00 (or \$0.95) and then choosing the item in the remaining periods.

4.2. Diagnostic Treatments. In addition to the three main treatments, we conducted three diagnostic treatments, two of which we describe here. We provide the details of the third diagnostic treatment, which provides a robustness test and directly ascertains whether subjects considered contingencies that would justify early exits in the OSP-E mechanism, in Section 5.2.1 after presenting the main results.

The first diagnostic treatment, No-Exit-OSP, is identical to OSP except that the subject is prevented from exiting the task when they choose ‘Exit the Market’ (NE refers to ‘no exit’ because after a subject exits the market, they cannot exit the task). Instead, we inform subjects that they will have to reconfirm their choice to exit for every lower clock-price until it reaches the randomly chosen price. The purpose of the No-Exit-OSP treatment is to see how robust any difference between OSP and UJS is to the possibility that subjects in the OSP treatment exit early simply because of opportunity costs, something which they cannot do in either the UJS or No-Exit-OSP treatments.

¹³Ex post, an upper bound of \$1.50 does not seem to dramatically change behavior in BDM: the fraction of rational valuations we find is very similar to Cason and Plott (2014) who used an upper bound between \$4 and \$8 for a \$2 item.

¹⁴Other UJS mechanisms are possible, but we use the one most similar to OSP-E for comparability. For example, an alternative UJS mechanism is one in which the clock-price increases each period, keeping everything else the same.

¹⁵For example, if a subject selected ‘Try to Sell’, a pop-up stated: “If the computer buys your item, you will receive the dollar amount of \$x.”

The second diagnostic treatment addresses the second goal of the experiment: to understand whether the existence of justifiable dominated actions leads to failures of the OSP-E mechanism (Hypothesis 3). This treatment uses the diagnostic UJ-OSP-D mechanism from Example 4. This mechanism is essentially identical to the OSP-E mechanism except that subjects were forced to choose the item when the price reaches \$1.00, a fact which we emphasized in the instructions. By forcing subjects to choose rationally at \$1.00, choosing the price becomes the uniquely justifiable strategy. Comparing the relative performance of this mechanism and the OSP-E mechanism identifies the importance of justifiable actions.¹⁶

4.3. Diagnostic Questions. In all four dynamic mechanism treatments (OSP, No-Exit-OSP, UJS, and SOSP), we asked subjects a single incentivized question (\$0.50 for correct answer) after the main elicitation task.

If the subject had chosen the item at some clock-price, we reminded them of the highest clock-price at which they had chosen the item, and then asked them for the highest payoff they could have received if they had instead chosen ‘Try to Sell’ in that period. The question tested subjects’ understanding of the payoff implications of exiting the mechanism early.

If the main task ended before the subject had ever chosen the item, we instead asked for the clock-price at which they should first choose the item to maximize earnings.¹⁷ In this way, we obtain a measure of rationality for subjects that are censored by the mechanism. However, because of difficulties interpreting subject responses to this incentivized question relative to valuations elicited via the mechanism, our analysis focuses on results obtained directly through the mechanisms.¹⁸

4.4. Logistical Details. All five treatments were run on the Prolific platform in August-September of 2021 (BDM, OSP, and UJ-OSP) and July-August of 2022 (UJS and No-Exit-OSP). We recruited between 297 and 301 subjects located in the U.S. for each treatment. Average earnings were \$2.47, or about \$18.39/hour. Subjects first read instructions about their assigned mechanism and then answered a series of non-incentivized quiz questions to assess their comprehension of the instructions. We did not prevent the participation of subjects who failed to answer the quiz questions correctly. The full sets of instructions are reported in Appendix E.

¹⁶For completeness, we also ran a treatment using the UJ-OSP-D mechanism in which subjects had to reconfirm their choice to exit for all prices, as in No-Exit-OSP. We discuss the results of this No-Exit-UJ-OSP treatment in Table B.2 of Appendix B.

¹⁷Of course, this question is only meaningful in an induced valuation environment .

¹⁸For completeness, Table B.1 of Appendix B compares behavior obtained through the mechanism with that obtained via the questions, showing fairly consistent results.

5. EXPERIMENTAL RESULTS

In Section 5.1, we summarize the main results and show that all of the dynamic mechanisms outperform BDM. In Section 5.2, we show that justifiable dominated actions lead to failures of the OSP-E mechanism. Finally, in Section 5.3, we compare the UJS and OSP treatments.

5.1. BDM versus Dynamic Mechanisms.

5.1.1. *Samples.* In our analysis, we report results for the Full sample, but also for subsamples of subjects that answered all quiz questions correctly in each treatment (Attentive sample). Subjects that do not answer the questions correctly are likely to introduce noise due to lack of attention to the instructions. In all of our histograms and tables, we refer to the switching points obtained under the OSP-E, UJS-E, and UJ-OSP-D mechanisms as the elicited *valuations*.

5.1.2. *Dealing with Censoring.* Although the dynamic mechanisms (OSP-E, UJS-E, and UJ-OSP-D) elicit truthful valuations for each individual, subjects who switch earlier at higher prices are less likely to be censored by the mechanism, leading to an upwards bias in the *distribution* of observed valuations (or switching points).¹⁹ We remove this bias by trimming both the Full and Attentive samples of subjects to include only subjects for whom the randomly chosen clock-price was less than \$1.00, ensuring rational valuations are neither censored nor under-represented in this subsample. In these samples, we also include all subjects that were censored at valuations less than \$1.00, taking the last observed price as a subject's valuation if they were censored. In this way, we correctly observe the proportion of subjects with high ($> \$1.00$), rational ($\1.00 or $\$0.95$, due to indifference at $\$1.00$), and low ($< \1.00) valuations.²⁰

5.1.3. *Results.* Figure 5.1 provides histograms of the elicited valuations for the Full samples for all five treatments.²¹ For the UJS treatment, this figure excludes subjects that switched more than once between the price and the item because their valuation is then indeterminate.²²

To perform statistical comparisons across mechanisms, we begin by summarizing the data in Table 2. We separate valuations into rational ($\$1.00$ and $\$0.95$), those above $\$1.00$, and

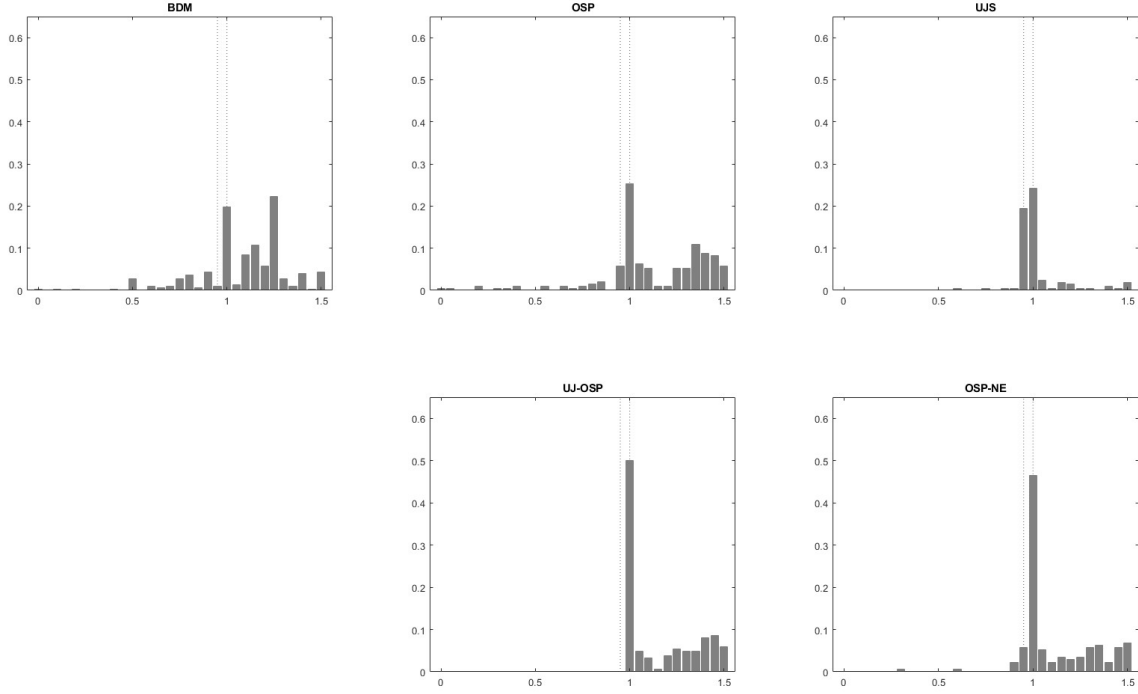
¹⁹For example, if half of subjects would have switched to the item at $\$1.50$ and half would have switched at $\$1.00$, then, because of censoring we would expect to observe $\frac{\frac{1}{2}}{\frac{1}{2} + \frac{2}{3} \cdot \frac{1}{2}} = \frac{3}{5}$ of subjects switching at $\$1.50$, thus overestimating the proportion of irrational choices. One-third of the rational subjects would be censored, on average.

²⁰The actual valuations for subjects with valuations less than $\$1.00$ will be overestimated in some cases, but, in practice, there are very few such observations and we do not use their exact values for any of our results.

²¹The bins are $\$0.05$ wide so that for the dynamic mechanisms none of the valuations are pooled.

²²We report details of the multiple-switching behavior in Section 5.3.

FIGURE 5.1. Valuations



Notes: Histograms (proportions) of valuations in all treatments. The dashed vertical lines indicate rational valuations (\$0.95 and \$1.00). Each panel is for a trimmed sample of subjects (those for which the randomly chosen clock-price was less than \$1.00). For the UJS treatment, subjects that switched multiple times are not shown and hence the proportions do not add up to 1. The UJ-OSP treatments uses the UJ-OSP-D mechanism that mechanically prevents valuations below \$1.00.

those below \$0.95. For the UJS treatment, we also report the frequency of multiple-switching behavior. For this treatment, all subjects that switched more than once are counted as not having reported rational valuations in all statistical tests. We report the data for the Full and Attentive samples of subjects. N indicates the size of each sample.

With the exception of the BDM treatment, we see considerably more rational behavior in the Attentive samples, consistent with inattention among those who do not correctly answer the quiz questions. The fact that understanding the quiz questions leads to better decisions in the OSP-E, UJS-E, and UJ-OSP-D mechanisms, and not in the BDM mechanism suggests that BDM fails to satisfy Behavioral Incentive Compatibility (Danz, Vesterlund, and Wilson (2022)): better comprehension of the incentives actually decreases truthful reporting.

TABLE 2. Summary of Valuations

Treatment	Sample	N	Rational Valuations	Valuations >\$1	Valuations <\$0.95	Multiple Switch
BDM	Full	297	19.9%	62.0%	18.2%	-
	Attentive	102	15.7%	60.8%	23.5%	-
OSP	Full	193	31.1%	57.5%	11.4%	-
	Attentive	60	40%	43.3%	16.7%	-
UJ-OSP	Full	186	50%	50%	-	-
	Attentive	45	77.8%	22.2%	-	-
No-Exit-OSP	Full	174	52.3%	44.3%	3.5%	-
	Attentive	67	68.7%	25.3%	6.0%	-
UJS	Full	206	43.7%	10.7%	1.9%	43.7%
	Attentive	74	56.8%	8.1%	5.4%	29.7%

Notes: Fraction of subjects with rational valuations and those above and below \$0.95 for each treatment. For all mechanisms, we report the results separately for the subsamples of subjects that answered all quiz questions correctly (Attentive). The UJ-OSP-D mechanism in the UJ-OSP treatment is a diagnostic mechanism that prevents valuations below \$1.00.

We begin our assessment of the relative performance of the mechanisms by comparing the percentage of rational valuations in the dynamic mechanisms to BDM. Both UJS-E and OSP-E (whether in treatment OSP or No-Exit-OSP) vastly outperform BDM in the fraction of subjects reporting a valuation in the range \$0.95-\$1.00. To make statistical comparisons, we use two-sided Fisher exact tests to compare the relative proportions. Here, the largest p-value of the three comparisons is that for OSP ($p = 0.005$). The improvements in rational decisions are also large in magnitude, increasing from 19.9% in BDM to at least 31.1% in the dynamic mechanisms. If we filter on the subsets of attentive subjects that answered the quiz questions correctly, the improvements are even larger: increasing from 15.7% in BDM to at least 40%.

Result 1: *The OSP-E and UJS-E mechanisms greatly outperform BDM in eliciting truthful valuations, supporting Hypotheses 1.i and 2.i.*

The poor performance of BDM is not unexpected given previous findings. In the closest related studies, Cason and Plott (2014) find 16.7% of inexperienced subjects' valuations are rational, while Martin, Muñoz-Rodríguez (2022) find 17.4%. Both of these results are similar to the 15.7% we find.²³

The better performance of the OSP-E mechanism provides evidence, in addition to Li (2017a), that OSP mechanisms in general outperform their more traditional counterparts.

²³Previous studies treated \$0.95-\$1.05 as rational instead of \$0.95-\$1.00. If we use the wider range, 19.6% of bids would be rational, which is slightly higher than in the previous studies, perhaps because we explicitly told subjects that bidding \$1.00 is optimal. Martin and Muñoz-Rodríguez (2022) ran a version of BDM which asked subjects for their minimum selling price, while reframing each contingency as a different computer bidder. This reframing increased the frequency of rational choices to 34.9%.

However, between 30% and 70% (depending on the treatment and sample) of subjects that understand the quiz questions still fail to report rational valuations in the OSP-E mechanism.

5.2. Uniquely Justifiable Strategies. Our hypothesized reason for failures of the OSP-E mechanism is that both actions are justifiable (Hypothesis 3): subjects may choose the item at clock-prices greater than \$1.00, because they fail to understand that rational play will lead them to exit at \$1, thus avoiding all contingencies in which they can receive less than \$1.00. We test this hypothesis by comparing behavior across the OSP-E and UJ-OSP-D mechanisms.

Recall that the UJ-OSP-D diagnostic mechanism is *identical* to the OSP-E mechanism except that subjects were forced to keep the item when the price reached \$1.00, thus preventing them from earning less than \$1.00. If the fact that both actions are justifiable is driving early exits in the OSP-E mechanism, we would expect more valuations above \$1.00 in OSP-E than in UJ-OSP-D. Consistent with Hypothesis 3, this is what we observe: 57.5% versus 50% in the Full sample. We can reject the null of no more high valuations in OSP-E at the 10 percent level ($p = 0.086$).²⁴ If we filter on attentive subjects that correctly answer the quiz questions, the treatment effect is even larger and highly significant: the fraction of valuations above \$1.00 decreases from 43.3% in OSP-E to 22.2% in UJ-OSP-D ($p = 0.019$).²⁵

Some subjects, rather than being concerned about receiving less than \$1.00, may exit early for other reasons, such as inattention. These subjects would add noise to both the UJ-OSP and OSP treatments. To identify subjects that are intentionally forgoing the possibility of a higher payoff due to a fear of receiving a low payoff, we use the diagnostic question (section 4.3). Specifically, we compare the fraction of subjects that exit early but also correctly answer the incentivized question about the highest possible payoff they could have gotten by instead choosing ‘Try to Sell’. We find that the fractions of such subjects in each treatment are 37.3% (OSP) and 28.0% (UJ-OSP), a difference that is statistically significant ($p = 0.034$).²⁶ This evidence, together with the fact that we find significant differences between the OSP and UJ-OSP treatments among subjects that answer all of the quiz questions correctly, provides evidence that the justifiability of exiting early causes some subjects to do so.

What makes this empirical result particularly striking is that the OSP-E mechanism is One Step Simple (Pycia and Troyan, 2022) – even under the simple provisional plan of exiting one step later, the worst outcome from following the dominant strategy at the current step

²⁴Hypothesis 3 is one-sided - that UJ-OSP-D mechanisms result in less high valuations than OSP-E mechanisms - and hence we use one-sided Fisher exact tests.

²⁵In comparing OSP and UJ-OSP when filtering on the quiz questions, one might be concerned about differential selection. However, the instructions differ in only the sentence indicating one must choose the item at \$1.00 in the UJ-OSP-D mechanism. The quiz questions are identical.

²⁶We interpret these results with some caution - it could be that subjects come to understand the payoff consequences of their decision to choose the item only after being asked the question, or that answering the diagnostic question correctly also serves to filter on more attentive subjects.

does better than the best outcome from any deviation. In Pycia and Troyan (2022)’s set of nested simplicity standards for obviously dominant mechanisms, OSP-E is as simple as possible given that an SOSP mechanism does not exist. And, yet, even in this case, subjects choose dominated actions that are justifiable.

5.2.1. Direct Evidence for Lack of Foresight in OSP mechanisms. We designed an additional treatment, OSP-L, that is very similar to the OSP treatment, but (i) assesses the robustness of the early exits (high valuations) observed with the OSP mechanism to several differences, and (ii) provides direct evidence of subjects considering contingencies that justify early exits. First, rather than eliciting the value of \$1, in OSP-L we elicit the value of a lottery that pays \$1.20 or \$1.00 with equal probability. Although using an object of known and riskless valuation (\$1.00) has the advantage of being able to sharply identify rational behavior, and makes our results directly comparable to those in the literature (Cason et. al., 2006; Li, 2017), it may feel artificial to some subjects. Second, we had subjects repeat each task three times to allow for learning. Third, we increased the stakes relative to the previous treatments. Subjects earned an average of \$4.13, or about \$27.85/hour, which is significantly higher than the median stakes in lab or online experiments.²⁷

Fourth, and perhaps most importantly, we directly asked subjects *after* the three repetitions the diagnostic yes/no question, “*Were you concerned that you might sell the lottery ticket for less than \$0.90?*” Although not incentivized, the question provides direct evidence of concerns about receiving a low payoff, and allows us to compare behavior across subjects with and without such concerns. In particular, subjects that self-report such concerns should be more likely to exit early at a high valuation, providing complementary evidence to that of the previous section.

We classify choices as rational if the elicited valuation (exit price) of a subject is between \$1.00 and \$1.15 (inclusive), allowing for a certainty equivalent strictly between \$1 and \$1.20. Thus, using lotteries in place of a \$1 item makes our classification more restrictive: subjects who value the lottery at \$1.01, but irrationally justify exiting at \$1.10 or \$1.15 will be classified as rational.

Table 3 provides a summary of the results for OSP-L, corresponding to Table 2. As before, in each repetition, we include only subjects for whom the randomly chosen clock-price is less than \$1.00. The three numbers indicate the number of subjects and fraction of each type of behavior in each repetition.²⁸ The proportion of subjects who exit early (at or above \$1.20) in the third round of OSP-L are 57.7% and 42.4% in the Full and Attentive samples. In comparison, the proportion of subjects who exited early in OSP (when valuing the \$1 item) are almost identical: 57.5% and 43.3% in the Full and Attentive samples. Thus, even though

²⁷The treatment was run on Prolific in February of 2023. 303 subjects participated in total.

²⁸The number of subjects varies across repetition due to differences in the random ending prices.

TABLE 3. Valuations - OSP-L Robustness Treatment

Sample	N			Rational Valuations			Valuations > \$1.15			Valuations <\$1.05		
Full	208	200	175	31.7%	31.0%	24.6%	58.2%	55.0%	57.7%	10.1%	14.0%	17.7%
Attentive	76	71	66	46.1%	56.3%	44.0%	47.4%	37.5%	42.4%	6.6%	6.9%	13.6%

Notes: Fraction of subjects with rational valuations (exit price), and those above \$1.15 and below \$1.05 for each treatment. We report the results separately for the subsamples of subjects that answered all quiz questions correctly (Attentive). The results are reported separately for each repetition, in order.

we made several changes that should have increased rational behavior (repetition, larger incentives, and the use of a different item), exiting at high valuations is just as common as in our main treatments.²⁹

To provide direct evidence of concerns about low payoffs, we turn to the diagnostic question that asked subjects if they were concerned that they might receive less than \$0.90.³⁰ To condition on attentiveness or comprehension, we focus on the 37.3% of subjects that answered all five quiz questions correctly. Among them, 48.0% report concerns about a contingency in which they receive less than \$0.90. Even though we are filtering for attentiveness, given the binary nature of this question, this is what we’d expect if filtered subjects were somehow inattentive to this question and answering it randomly. Critically, however, the fraction of early exits is 48.4% among those that report concerns versus 31.2% among those that don’t, a difference that is statistically significant with a two-sided Fisher exact test ($p = 0.040$). Thus, those that self-report concerns are also much more likely to act on those concerns.³¹

We summarize the results of the UJ-OSP and OSP-L treatments as follows.

Result 2: *Subjects consistently make more mistakes - choose the item at clock-prices greater than a rational subject would - in OSP mechanisms that require correctly forecasting future actions than in diagnostic mechanisms that do not. Furthermore, attentive subjects that consider low payoff contingencies more often take dominated actions that are justifiable in those contingencies (exit early).*

The reduction in high valuations in UJ-OSP-D can be interpreted in two different ways. Subjects may have mistaken beliefs about their future actions (or not have well-specified

²⁹To see if forcing subjects to exit at \$1.00 in the UJOSP-D treatment makes it salient to exit at exactly that price instead of an earlier price, in the OSP-L treatment, we forced subjects to exit at \$0.05. We find no evidence that the forced exit price is driving behavior: we observe only 3 exits at that price across all subject-repetitions (versus one in the OSP treatment).

³⁰In the following results, we consider the subsample of subjects for whom the random ending price was less than \$1.20 (rather than \$1.00), because this is the largest subsample for which we can identify the proportion of exits at and above \$1.20, thus maximizing statistical power. The results are very similar, but statistically weaker if we instead use a random ending price of \$1.00 or less.

³¹Among inattentive subjects that answer at least one quiz question incorrectly, 59.3% report concerns about receiving less than \$0.90 but these concerns do not drive behavior: the fraction of early exits is 63.3% among those that report concerns versus 67.0% among those that don’t ($p = 0.560$). Thus, inattention does not drive these concerns, but instead introduces noise.

beliefs at all). Or, subjects may have *correct* beliefs - they rationally choose to exit because they correctly forecast that they will make mistakes in the future. That is, rather than not thinking through their future actions, subjects know the optimal strategy but are instead worried that they might make simple mistakes in implementing it.

Several factors favor the first interpretation. First, simple mistakes from clicking the wrong action should be rare given that subjects have to confirm each choice that they make. Second, in Appendix D, we simulate a subject’s decision to continue when they rationally expect to make a mistake with a constant probability in each subsequent period. There, we show that for any probability of mistake between 0 and 1, the expected value of continuing exceeds that of exiting (\$1.00). Intuitively, for a very low probability of mistake, one should continue because it is likely one will exit at \$1.00 so that getting less than \$1.00 is very unlikely. But, with a very high probability of mistake, it is likely one exits before ever reaching \$1.00 which also makes getting less than \$1.00 unlikely. Lastly, in Chakraborty and Kendall (2025), we provide strong evidence of incorrect or ill-formed beliefs about one’s own future actions in a simple two-period investment problem.

5.3. UJS Results. Given that some failures of the OSP-E mechanism appear to be due to both actions being justifiable, we turn to the question of whether or not the UJS-E mechanism does better (Hypothesis 2.ii).

In terms of rational behavior, the answer depends upon which treatment we compare UJS to. Compared to the more natural OSP treatment, which mirrors the experiments from Li (2017a) and Breitmoser and Schweighofer-Kodritsch (2021), and in which exiting ends the task, the UJS-E mechanism does better ($p = 0.001$). But, compared to the diagnostic No-Exit-OSP treatment where we force subjects to click through all clock-prices even after exiting, the UJ-E mechanism does slightly worse (significant at the 10% level ($p = 0.100$)).

Although we intended the No-Exit-OSP treatment to remove early exits in the OSP mechanism that are due to subjects rushing to get through the experiment faster, in hindsight it appears to have had the unintended effect of making subjects think more carefully about when to exit, leading to more rational behavior independent of the opportunity cost channel. As evidence, we observe a significant *decrease* in the fraction of late exits (valuations < 1) when moving from OSP to No-Exit-OSP ($p = 0.005$), a pattern that cannot be explained by a general shift towards subjects exiting later in No-Exit-OSP.

Even though the No-Exit-OSP treatment appears to have nudged subjects to better think through the mechanism, we don’t want to draw strong conclusions about which mechanism induces more rational behavior as it depends on the precise implementation details.

Result 3: *The UJS-E and OSP-E mechanisms induce similar proportions of rational behavior.*

When we look at valuations above \$1.00, a very different picture emerges - there are significantly less such valuations in UJS than OSP or No-Exit-OSP ($p < 0.001$). For valuations less than \$0.95, the rates are very low and comparable across treatments.

Given significantly fewer high valuations and similar low valuations, we would expect the UJS-E mechanism to significantly outperform the OSP-E mechanism. However, the UJS-E mechanism allows for a form of behavioral mistake that the OSP-E mechanism does not – switching between the price and the item multiple times. As in the large literature using multiple price lists for elicitation, we find multiple switching to be common, with 30-45% (depending on the sample) of subjects switching multiple times. In fact, multiple switching is by far the most common form of failure of the UJS-E mechanism.

Figure B.1 in Appendix B plots the choices of all subjects that switch more than once. Of the ninety subjects that switch multiple times, twelve subjects make only a single mistake. At the other extreme, some subjects make mistakes more often than not. Chew et al. (2022) show that ‘regular’ switching – starting by choosing the price and ending with choosing the item – is consistent with deterministic choice under non-expected utility preferences and that ‘non-regular’ switching can only be explained by stochastic choice. In our results, 53.3% of the multiple-switch behavior is regular so our results don’t shed light on whether the behavior is deliberate or not.³²

The UJS treatment also allows us to test directly whether obvious dominance (OD) improves behavior beyond the simplification already afforded by the UJS-E mechanism. Above \$1.00, choosing the price is not OD, while in the continuation game below \$1.00, choosing the item is. In both cases, the respective actions are uniquely justifiable in the simplified game. To isolate the effect of OD, we compare the fraction of OD-consistent choices in the range 1.50–1.05 to those in 0.95–0.50, such that the ranges are of equal size and the costs of deviation are identical.

We find that 81% of choices are OD-consistent in the upper range, compared to 85% in the lower range. A linear regression of an OD-consistent choice indicator on a dummy for the price range shows this difference is statistically significant ($p = 0.020$). However, once we include subject fixed effects, the difference vanishes ($p = 0.162$) and the coefficient is very

³²Other experiments have found evidence of deliberate randomization when the offered choices involve objective lotteries (Agranov and Ortoleva (2017); Agranov, Healy, and Nielsen (2023); Dwenger, Kübler, and Weizsäcker (2018)), ambiguity (Cettolin and Riedl (2019)), time preferences (Agranov and Ortoleva (2017)), social preferences (Sandroni, Ludwig, and Kirchler (2013); Agranov and Ortoleva (2017); Miao and Zhong (2018)), and even choices with dominated options (Rubinstein (2002)). Agranov and Ortoleva (2023) find that when comparing a fixed amount $\$x$ with a lottery that pays \$20 or \$0 with equal chance, subjects typically randomize for all x ’s between \$5.3 and \$12, indicating that deliberate randomization is not constrained to knife-edge cases. On the other hand, recent work with multiple price lists has shown that nudges can reduce multiple switching behavior (Yu, Zhang, and Zuo (2021)), suggesting that at least for some subjects the behavior may instead be due to mistakes.

close to zero (0.0000).³³ We conclude that OD provides, at most a very small improvement in behavior in the UJS-E treatment.

6. DISCUSSION

In this paper, we introduced the concept of *uniquely justifiable strategies*. A mechanism is UJS if it does not require contingent reasoning because best-responding to any contingency leads to the dominant strategy. Among binary allocation mechanisms, we fully characterize the set of UJS mechanisms providing a foundation for the use of price-list structures. We then show how the UJS concept is useful in explaining existing evidence on performance differences across mechanisms in a wide variety of settings.

Using an experiment, we show that the OSP-E and UJS-E elicitation mechanisms greatly outperform standard BDM, and that the OSP-E mechanism’s performance is limited by players considering contingencies outside of optimal future actions. From a practical standpoint, our results highlight two simple elicitation mechanisms that outperform BDM. More importantly, they suggest a broader design principle: mechanisms that provide uniquely justifiable strategies can potentially achieve better outcomes.

Our results comparing OSP-E and UJS-E point to a fundamental tradeoff in the design of robust mechanisms. In many practical applications, the OSP property requires moving from a static mechanism (e.g, BDM) to a dynamic mechanism (e.g, OSP-E) that intuitively makes it simpler to understand each decision. But, the dynamic nature of such OSP mechanisms requires rational foresight of one’s future actions, which some subjects do not appear to have. The benefit of uniquely justifiable strategy mechanisms, as in an MPL, is that they eliminate the need for rational foresight by creating a game structure in which current actions can be taken independently of future actions. Unfortunately, the benefit of this structure may also be its weakness – subjects seem averse to, or incapable of, making consistent decisions across different rows of a price list. To design more robust mechanisms, we then need to better understand the root of inconsistent choices and the means of preventing them.

As with any empirical test of theory, we are only able to jointly test the theory and a series of implementation decisions. As such, future work should stress test our results, in particular by implementing the mechanisms with other subject pools and implementation details (instructions, etc.). Brown, Liu, and Tsoi (2023) provide some such evidence, comparing BDM, OSP-E, and UJS elicitation mechanisms using subjects in a laboratory. In their experiments, the UJS-E mechanism outperforms both BDM and OSP-E, which appears to be primarily due to lower rates of multiple switching behavior (under 20%) than we observe. Future work can also explore the empirical performance of some of the other UJS mechanisms we propose, such as the UJS alternative to the VCG mechanism.

³³Logit regression results are very similar.

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APPENDIX A. DEFINITIONS AND PROOFS

A.1. Visibly Different Strategies. Let $t(S_p, S_{-p})$ be the set of all histories player 1 reaches when 1 plays S_p and others play S_{-p} . Let S^i and S^j be weakly dominant strategies for p 's types v^i, v^j respectively.

Definition 7. S^i and S^j are *visibly different* under S_{-p}^* if there exists a common history $h_{ij, S_{-p}^*} \in t(S^i, S_{-p}^*) \cap t(S^j, S_{-p}^*)$ where $S^i(h_{ij, S_{-p}^*}) \neq S^j(h_{ij, S_{-p}^*})$, and, (ii) for $k = i, j$,

$\arg \max_{a \in \mathcal{A}(h_{ij, S_{-p}^*})} u_1(h_{ij, S_{-p}^*}, a, S, S_{-p}^{**}, k) = \{S^k(h_{ij, S_{-p}^*})\}$

for all $S \in \mathcal{S}_1$ and some $S_{-p}^{**} \in \mathcal{S}_{-p}$ — comprising of one weakly dominant strategy for some type of each other player, together with Nature's move.

For example, take OSP-E, and let S^1 and $S^{.95}$ be the dominant strategies of two types that exit at all clock-prices below their valuations of \$1.025 and \$.975 respectively. These strategies are visibly different. Fix any $S_{-1}^* = v^r \leq 1$, and fix h_{ij, S_{-1}^*} as the common history reached by both strategies at the clock-price of \$1. Clearly, $S^1(h_{ij, S_{-1}^*}) \neq S^{.95}(h_{ij, S_{-1}^*})$. Furthermore, for $S_{-1}^{**} = v^r = 1$, both types are playing their unique best responses at h_{ij, S_{-1}^*} , irrespective of what S they play in the rest of the game. The following example illustrates why part (ii) of the definition is required.

Example 8. Consider the one-player posted-price mechanism which randomly draws a $v^r \in \{0, .05, \dots, 1.50\}$ and offers the player a choice between the item and v^r . Each possible draw of v^r is a different information set, and each type, $v^k \in \{.025, .075, \dots, 1.525\}$, has a strongly obviously dominant strategy of choosing the item if and only if $v^k \geq v^r$. Thus, this mechanism satisfies (i) of Definition 1, but not (ii) because at no S_{-1}^* is there a common history where one observes any pair of types below (or above) v^r playing differently. Definition 7 rules out such cases.

By part (ii) of Definition 7, there must exist a contingency $(h_{ij, S_{-p}^*}, S, S_{-p}^{**})$ to which $S^k(h_{ij, S_{-p}^*})$ is the unique best response for $k = i, j$, and thus, i and j 's dominant strategies *must* play differently at h_{ij, S_{-p}^*} .³⁴ S_{-p}^{**} can depend on i, j, S_{-p}^* , and it need not be equal to S_{-p}^* .

A.2. SOSP infeasibility (Proposition 1).

Proof. (Part i) Suppose such a separating SOSP mechanism G exists for player $p = 1$ (WLOG). We will show a contradiction in 4 steps.

³⁴This restriction is important, as otherwise the mechanism from the previous example can be superficially fixed without aiding identification. Add an action for Nature, S_{-1}^* , where irrespective of the player's actions at the succeeding information sets, every resulting outcome is $\bar{o}$. As 1's actions are irrelevant to payoffs under S_{-1}^* , any action can be supported at the information sets succeeding S_{-1}^* as a weakly dominant strategy for any type. In this case, the $S^i(h_{ij, S_{-1}^*}) \neq S^j(h_{ij, S_{-1}^*})$ condition can be trivially satisfied, without types having strict incentives for playing those actions.

Step 0: Pycia and Troyan (2023) show that any SOSP mechanism takes the form of a sequential choice game³⁵, where each player makes at most one choice (Theorem 8). In eliciting the valuation from a single player 1, this restriction implies that any SOSP mechanism offers a single payoff-relevant choice to player 1. Further, to meet the visibly different strategies requirement of Definition 1, there must exist an S_{-1}^* and a common history h^* reached under S_{-1}^* , such that at h^* , every type i takes a different action, a_i . Let the set of possible utility values, as a function of valuation v , from action a_i be given by $\Pi_{a_i}^G(v) \subseteq \Pi(v) \equiv \{\pi(v) : \pi(v) = pv + r, p \in \{0, 1, 2, \dots, m\}, r \in \mathbb{R}\}$. For example, if action a_i leads to outcomes of (1 item + \$10) or (\$20), then, $\Pi_{a_i}^G(v) = \{v + 10, 20\}$.

Step 1: (Editing the SOSP mechanism so that each SOD action leads to a unique outcome) Let there exist some type i , for whom the SOD action, a_i , has $\Pi_{a_i}(v)$ with more than one utility value. This is only possible when the action a_i can lead to multiple possible outcomes. We modify the outcomes at the terminal nodes succeeding a_i by replacing all the outcomes by i 's favorite outcome from a_i , such that $\Pi_{a_i}^G(v)$ becomes

$$\Pi_{a_i}^{G'}(v) = \arg \max_{\pi \in \Pi_{a_i}^G} \pi(v^i)$$

in a new game $\hat{G}$. That is, in $\hat{G}$, action a_i always leads to the outcome that provided the highest utility to type i in G . For example, if the action, a_i pays outcomes (1 item + \$10) or (\$20), and is used to separate a type with valuation $v = \$15$, then, this step would modify the action to now pay (1 item + \$10) as the unique outcome at all terminal nodes succeeding a_i . We claim that this creates a new SOSP mechanism that still separates all the types. For type i , action a_i is still SOD at h^* :

$$\min \Pi_{a_i}^{\hat{G}}(v^i) \geq \underbrace{\min \Pi_{a_i}^G(v^i) \geq \max \Pi_{a_j}^G(v^i)}_{\text{because the action was previously SOSP}} = \max \Pi_{a_j}^{\hat{G}}(v^i) \quad \forall j \neq i$$

and action $k \neq i$ remains not SOD for type i , as $\min \Pi_{a_k}^{\hat{G}}(v^i)$ and $\max \Pi_{a_i}^{\hat{G}}(v^i)$ are unchanged. Finally, the visibly different strategies requirement still holds. For all other types, the proof follows similarly.

Step 2: Applying Step 1 recursively for each type's separating action creates a final SOSP mechanism (say G^*) where $\Pi_{a_i}^{G^*}(v)$ is a singleton for every separating action a_i , and the visibly different strategies requirement holds at h^* .

Step 3: Given $m < n$, there must exist at least two types $x, y \in \mathbb{N}$ such that their separating actions, a_x and a_y , allocate an identical number of items. Let the utilities from those separating actions be $\Pi_{a_x}^{G^*}(v) = pv + r_x$ and $\Pi_{a_y}^{G^*}(v) = pv + r_y$. Assume $r_x > r_y$. Then,

³⁵A sequential choice game is a perfect-information game in which Nature moves first, if at all. The agents then move sequentially, with each agent called to play at most once.

both types would take action a_x because $px + r_x > px + r_y$. If $r_x < r_y$, then both types take action a_y . If $r_x = r_y$, both types are indifferent between each other's actions, which violates the visibly different strategies requirement. Thus, in all three cases, we have a contradiction.

Part ii: Let (k, m) denote the outcome with k items and $\$m$. Pick n actions $(b_i)_{i=1}^n$ such that b_n pays $((n-1), 0)$ and b_i pays $((i-1), \sum_{j=i}^{n-1} \frac{v_{j+1} + v_j}{2})$ for $i \in \{1, \dots, n-1\}$. Action b_i separates (v^i) when we ask the player to choose one of the n actions. We skip the calculations for brevity. $\square$

A.3. Characterization of UJS Mechanisms. We first introduce notation and assumptions about preferences. Item and non-outcomes can generally be written as a vector of item and money allocations that belong in $\{0, 1\} \times \mathbb{R}_+$. For simplicity, we will assume throughout that the player has quasilinear preferences. Thus, for outcomes, $o = (x, m)$ and $o' = (y, m')$, where $x, y \in \{0, 1\}$ and $m, m' \in \mathbb{R}_+$ one has $u_1(o, v^i) \geq u_1(o', v^i)$ if $xv^i + m \geq yv^i + m'$. It is easy to see that $x \geq y, m \geq m'$ with strict inequality for at least one of them implies $u_1(o, v^i) > u_1(o', v^i) \forall v^i$: in this case we say that o dominates o' . We say that an outcome $(x, m) \equiv z_1(h, aS_1, S_{-1})$ is undominated at (h, S_1, S_{-1}) if there does not exist some other outcome $(x', m') \equiv z_1(h, bS_1, S_{-1})$ at the same contingency (h, S_1, S_{-1}) such that $x' \geq x, m' \geq m$ with strict inequality for at least one of item or money components.

Proof. (Proposition 3) (i) We proceed by contradiction, for the case of item outcomes. The other case follows similarly. For notational simplicity, WLOG, we focus on player $p = 1$. Fix an action $a \in \mathcal{A}(I_1)$ and suppose there exist two payoff-relevant contingencies at I_1 ,

$$(h, S_1, S_{-1}) \quad \text{and} \quad (h', S'_1, S'_{-1}),$$

such that:

- $z_1(h, aS_1, S_{-1})$ is an undominated item outcome, but
- $z_1(h', aS'_1, S'_{-1})$ is *not* an undominated item outcome.

We consider two exhaustive cases.

Case 1: $z_1(h', aS'_1, S'_{-1})$ is a dominated outcome.

Then no type can justify playing a under (h', S'_1, S'_{-1}) . However, since the $n \geq 2$ types do not all have identical uniquely justifiable actions, and since a yields an undominated item outcome at (h, S_1, S_{-1}) , a must be justifiable for at least one type, say v^* , at (h, S_1, S_{-1}) . But if a is dominated at (h', S'_1, S'_{-1}) , then v^* must have a different justifiable action at (h', S'_1, S'_{-1}) . This contradicts the assumption that v^* has a uniquely justifiable strategy in the simplified game G' . Contradiction.

Case 2: $z_1(h', aS'_1, S'_{-1})$ is an undominated non-item outcome.

Let:

$$z_1(h, aS_1, S_{-1}) = (1, m_1), \quad z_1(h', aS'_1, S'_{-1}) = (0, m'_1).$$

At contingency (h, S_1, S_{-1}) : Let $b \neq a$ be the action such that $z_1(h, bS_1, S_{-1}) = (0, m_2)$ is the best undominated non-item outcome. Since $z_1(h, aS_1, S_{-1}) = (1, m_1)$ is undominated by assumption, and assuming quasilinear preferences, all types with:

$$v > m_2 - m_1$$

strictly prefer $(1, m_1)$ to $(0, m_2)$, and so will take action a as their uniquely justifiable action.

Types with:

$$v < m_2 - m_1$$

prefer the non-item outcome and take action b . Since $n \geq 2$ and not all types have identical justifiable actions, there exist types $v^1 > m_2 - m_1$ and $v^2 < m_2 - m_1$ whose uniquely justifiable actions are a and b , respectively.

At contingency (h', S'_1, S'_{-1}) : Let $c \neq a$ be an action such that $z_1(h', cS'_1, S'_{-1}) = (1, m'_2)$ is the best undominated item outcome. Then all types with:

$$v > m'_1 - m'_2$$

strictly prefer the item outcome and prefer action c , while types with:

$$v < m'_1 - m'_2$$

prefer the non-item outcome $(0, m'_1)$ and prefer action a . Hence, there exist types $v^3 > m'_1 - m'_2$ and $v^4 < m'_1 - m'_2$ who prefer actions c and a , respectively.

Now consider the overlap in types:

- If $m_2 - m_1 \geq m'_1 - m'_2$, then $v^1 > m_2 - m_1 \geq m'_1 - m'_2$, so v^1 prefers a at the first contingency but $c \neq a$ at the second. This contradicts uniquely justifiable actions.
- If $m_2 - m_1 < m'_1 - m'_2$, then $v^2 < m_2 - m_1 < m'_1 - m'_2$, so v^2 prefers $b \neq a$ at the first contingency but a at the second. This contradicts uniquely justifiable actions.

In either case, we find a contradiction. Therefore, action a must lead exclusively to item outcomes or exclusively to non-item outcomes across all contingencies.

(ii) The proof is by construction and proceeds in three steps.

Step 1: Constructing a truncated simplified game $(G')^t$ from the simplified game G' .

Apply the following truncation algorithm recursively to Player 1's information sets in G' , starting at an information set I that has no preceding history where 1 previously moved:

- **Scenario 1:** If all Player 1 types have the same uniquely justifiable action at an information set I , remove all other actions. If I has no payoff-relevant contingencies, remove all but one action.³⁶
- **Scenario 2:** If I does not satisfy Scenario 1, choose a payoff-relevant contingency c at I . Retain only the actions leading to the dominating item and non-item outcomes.³⁷

After operating on I , remove unreachable histories and proceed recursively to successors. After the algorithm has been applied on all information sets where 1 moves, we obtain the final truncated game, $(G')^t$.

Step 2: Show that justifiable strategies survive truncation.

- 2.1. Survival of uniquely justifiable strategies:** By construction, actions removed are not justifiable for any type. Hence all strategies that were uniquely justifiable in G' survive in $(G')^t$.
- 2.2. Truncation of future actions:** Truncating an action in a later information set might remove payoff-relevant contingencies from an earlier information set. Despite that, any strategy that was uniquely justifiable for some type v^j in G' , still plays a uniquely justifiable action at every information set that still retains a payoff-relevant contingency in $(G')^t$. And at least one such payoff-relevant contingency is retained.³⁸

Step 3: Constructing G^* .

Remove from G , the same edges (actions) and nodes (histories) removed to get $(G')^t$ from G' , yielding G^* . Construct the information partition in G^* so that it mirrors G , restricted to surviving histories. This is possible because any $h_1, h_2 \in P^{-1}(1)$ that belong to the same information set I in G , still retain identical actions. This in turn follows from how simplified games coarsen information sets.³⁹

$(G')^t$ is a simplified game of G^* and retains the two-action structure at each information set. By part (i) of the proposition, each action must lead exclusively to item or non-item outcomes. Thus G^* also inherits this structure.

³⁶We assume that all types play the same action in this case, and we keep that action.

³⁷At every contingency with a uniquely justifiable action, there has to be exactly one dominating outcome for each type of outcome (item/non item). For example, if there were two item outcomes that are identical and that dominate all other item outcomes, then there would no longer be a uniquely justifiable action for the type that chooses an action that leads to item outcomes. Further, we cannot have a Scenario 3 where all outcomes lead to the same type (item/non-item) of outcome, as then one outcome would dominate all other outcomes and this would already have been dealt with in Scenario 1.

³⁸In particular, by Definition A.1, the information set that contains history $h_{ij, S_{-1}}$ and guarantees S_i is visibly different than S_j in G , will retain a payoff-relevant contingency irrespective of which later actions get truncated.

³⁹Take two histories (nodes) $h_1, h_2 \in P^{-1}(1)$ that belong to the same information set I in G . As G' (weakly) coarsens 1's information sets in G , $h_1, h_2 \in P^{-1}(1)$ must also belong to the same information set, say I_p , in G' . Thus, when using the truncation algorithm on I_p in G' , any action removed from h_1 also must have been removed from h_2 .

Step 4: Show G^* is a UJS mechanism.

- **Dominant strategies preserved:** Any dominant strategy S^i for a type i in G remains in G^* as its corresponding uniquely justifiable strategy survived the truncation.
- **Visibility preserved:** Let S_{-1}^* be the strategy profile used to witness visible differences in G . The history h_{ij, S_{-1}^*} used to distinguish types i and j survives in G^* since it is on the path of both their justifiable strategies in G' and thus in $(G')^t$. The actions taken at h_{ij, S_{-1}^*} remain available and optimal at the relevant contingency.⁴⁰

Hence, G^* and $(G')^t$ together satisfy the UJS mechanism definition, complete the construction, and ensure that any UJS mechanism can be simplified into a price-list-like structure with binary choices. □

A.4. Mutual Exclusivity of OSP and UJS mechanisms. Li (2017a) shows that all OSP binary allocation mechanisms must have the form of personal clock auctions in which some *quitting* action can ‘clinch’ a *certain outcome* at every information set. We now show that personal clock auctions and UJS mechanisms are mutually exclusive in the world of binary allocation mechanisms. For simplicity, in our proof we assume that the certain outcome in the personal clock auction is an item outcome, but our proof applies with minimal changes if it’s a non-item outcome instead.

Proof. (Proposition 2) WLOG, let $p = 1$. For any (Player 1) strategy S' in the P1-simplified game, G' , let S be the corresponding unique strategy in G which leads to an identical set of terminal histories in G for every $S_{-1} \in \mathcal{S}_{-1}$. While every S' in G' must have a corresponding S in G , because of how G' coarsens the information partition, the opposite is not necessarily true.⁴¹

Suppose, for contradiction, that there exists a game G that is both a personal clock auction and a UJS mechanism, with $n \geq 3$ types. Pick three types i, j, k . By definition of a UJS mechanism, there must exist some S_{-1}^* such that the corresponding dominant strategies of all three pairs, S^j and S^k , S^i and S^j , and S^i and S^k are visibly different (definition 7) under some S_{-1}^* . Part (i) of Definition 7 implies that there exist jointly reached histories $h_{jk} \in t(S^j, S_{-1}^*) \cap t(S^k, S_{-1}^*)$, $h_{ij} \in t(S^i, S_{-1}^*) \cap t(S^j, S_{-1}^*)$, and $h_{ik} \in t(S^i, S_{-1}^*) \cap t(S^k, S_{-1}^*)$ with $S^j(h_{jk}) \neq S^k(h_{jk})$, $S^j(h_{ij}) \neq S^i(h_{ij})$, and $S^k(h_{ik}) \neq S^i(h_{ik})$. Further, for j and k , part (ii) of Definition 7 implies the existence of some S_{-1}^{**} such that for all $S \in \mathcal{S}_1$

$$(A.1) \quad \underbrace{\arg \max_{a \in \mathcal{A}^G(h_{jk})} u_1^G(h_{jk}, a, S, S_{-1}^{**}, v^j)}_{\neq} = \{S^j(h_{jk})\} \neq \{S^k(h_{jk})\} = \underbrace{\arg \max_{a \in \mathcal{A}^G(h_{jk})} u_1^G(h_{jk}, a, S, S_{-1}^{**}, v^k)}_{\neq}$$

⁴⁰Here we are using that G' (weakly) coarsens 1’s information sets in G , and thus the set of contingencies (and continuation strategies) in G' is a subset of those in G .

⁴¹The definition of UJS mechanisms only guarantees that the dominant strategies of every type in G have a corresponding uniquely justifiable strategy in G' .

By definition of a UJS mechanism, S^j , S^k , and S^i in G have corresponding uniquely justifiable strategies S'^j , S'^k , and S'^i in the simplified game G' . For any S' , $z_1^{G'}(h_{jk}, S'^j(h_{jk})S', S_{-1}^{**}) \neq z_1^{G'}(h_{jk}, S'^k(h_{jk})S', S_{-1}^{**})$, as any S' must have a corresponding S in G and $z_1^G(h_{jk}, S^j(h_{jk})S, S_{-1}^{**}) \neq z_1^G(h_{jk}, S^k(h_{jk})S, S_{-1}^{**})$ for any S using Equation (A.1).⁴²

Step 1 (h_{jk} , h_{ij} and h_{ik} cannot belong to the same information set in G):

Suppose h_{jk} , h_{ij} and h_{ik} are in the same information set I in G . We establish a contradiction. If the histories were in the same information set I in G , they would also be in the same information set I_p in G' .⁴³ Then, the three types must have different uniquely justifiable actions, a_1, a_2, a_3 , at I_p .⁴⁴ Fix any S_{-1}, S_1 and $h \in I_p$ for a payoff-relevant contingency in I_p . Define $o_x = z_1(h, a_x S_1, S_{-1})$ for $x \in \{1, 2, 3\}$. At least two of the three outcomes o_1, o_2, o_3 reached must be the same type (item/ non-item) of outcome. WLOG let these be o_1 and o_2 . If a_1, a_2, a_3 are uniquely justifiable actions, it must be that $u_1(o_1, v^1) > u_1(o_2, v^1)$ and $u_1(o_2, v^2) > u_1(o_1, v^2)$. These inequalities cannot hold simultaneously because all types must rank o_1 and o_2 identically.

Thus, at least two of h_{jk} , h_{ij} and h_{ik} must lie in two different information sets in G . Suppose, h_{jk} and h_{ij} belong to different information sets I_2 and I_1 in G , respectively. As $h_{jk} \in t(S^j, S_{-1}^*) \cap t(S^k, S_{-1}^*)$ and $h_{ij} \in t(S^i, S_{-1}^*) \cap t(S^j, S_{-1}^*)$, we know that $h_{jk}, h_{ij} \in t(S^j, S_{-1}^*)$. Thus, either $h_{jk} \succ h_{ij}$ and $I_2 \succ I_1$ or $h_{ij} \succ h_{jk}$ and $I_1 \succ I_2$. Assume $h_{jk} \succ h_{ij}$ and $I_2 \succ I_1$ WLOG. Let I_{p2} and I_{p1} be the pooled/ coarsened information sets in G' that contain h_{jk} and h_{ij} respectively.⁴⁵ As $h_{jk} \succ h_{ij}$, it must be that $I_{p2} \succ I_{p1}$.

Step 2 (Among types j and k who separate at h_{jk} , exactly one's (say j 's) dominant action at h_{jk} must be a quitting action):

As every S' in G' has a corresponding S in G , Equation (A.1) implies that at I_{p2} for all S'

$$(A.2) \quad \underbrace{\arg \max_{a \in \mathcal{A}^{G'}(h_{jk})} u_1^{G'}(h_{jk}, aS', S_{-1}^{**}, v^j) = \{S'^j(h_{jk})\}}_{\text{left side}} \neq \underbrace{\{S'^k(h_{jk})\} = \arg \max_{a \in \mathcal{A}^{G'}(h_{jk})} u_1^{G'}(h_{jk}, aS', S_{-1}^{**}, v^k)}_{\text{right side}}$$

Thus, $z_1^{G'}(h_{jk}, S'^j(h_{jk})S', S_{-1}^{**}) \neq z_1^{G'}(h_{jk}, S'^k(h_{jk})S', S_{-1}^{**})$, and neither outcome can be dominated by the other. Hence, these two must be different types of outcomes, and thus one of these two outcomes, say $z_1^{G'}(h_{jk}, S'^j(h_{jk})S', S_{-1}^{**})$ must be an item outcome.⁴⁶ As G

⁴²Again using the fact that G' weakly coarsens Player 1's information sets in G .

⁴³Particularly, because G' weakly coarsens 1's information sets.

⁴⁴Take any two types, say i and j . Part (ii) of Definition 7 (visibly different) implies that at the contingency $(h_{ij}, S', S_{-1}^{**})$ in G' , these two types must have different justifiable actions $S'^i(h_{ij})$ (that is identical to $S^i(h_{ij})$) and $S'^j(h_{ij})$ (that is identical to $S^j(h_{ij})$) respectively. But, given G is a UJS mechanism, those two actions must also be uniquely justifiable for i and j . Therefore, i and j must have different uniquely justifiable actions at I_p .

⁴⁵Because G' weakly coarsens 1's information sets.

⁴⁶If the certain outcome in the clock auction is a non-item outcome, then we would simply exchange item and non-item outcomes everywhere in the text here onwards.

is also a personal clock auction, all quitting actions in G (and hence in G') must lead to the same item outcome, say $(1, \bar{m})$.⁴⁷ Thus, $z_1^{G'}(h_{jk}, S'^j(h_{jk})S', S_{-1}^{**}) = (1, \bar{m})$. By equation A.3, for any deviating action $b \neq S'^j(h_{jk})$, j must prefer the outcome $z_1^{G'}(h_{jk}, bS', S_{-1}^{**})$ *strictly less* than $(1, \bar{m})$.

Step 3 (Establishing contradiction with j having a uniquely justifiable action (in G') at h_{ij}):

Choose S' such that it leads from h_{ij} to h_{jk} in G' and $S'(h_{jk}) = b$. Because equation A.2 holds for all S' , editing S' does not affect the relations in A.2.

$h_{jk}, h_{ij} \in t(S^j, S_{-1}^*) \cup t(S^k, S_{-1}^*)$ and $h_{jk} \succ h_{ij}$ imply $S^j(h_{ij}) = S^k(h_{ij})$ and hence $S'^j(h_{ij}) = S'^k(h_{ij})$.⁴⁸ The common action, $S'^j(h_{ij})$, taken at h_{ij} could not have been the quitting action (say $a^q \in \mathcal{A}^{G'}(h_{ij})$), as in that case, all following outcomes would be the item outcome, making it impossible for types j, k to separate thereafter at h_{jk} . Therefore, it must be that

$$(A.3) \quad u_1^{G'}(h_{ij}, S'^j(h_{ij})S', S_{-1}^{**}, v^j) = u_1^{G'}(h_{jk}, bS', S_{-1}^{**}, v^j) < u_1^{G'}(h_{jk}, S'^j(h_{jk})S', S_{-1}^{**}, v^j) = u_1^{G'}(h_{ij}, a^q S', S_{-1}^{**}, v^j)$$

where the first equality follows from the construction of S' (it leads to h_{jk} and $S'(h_{jk}) = b$), the inequality follows from equation A.2, and the last equality follows from $z_1^{G'}(h_{jk}, S'^j(h_{jk})S', S_{-1}^{**}) = z_1^{G'}(h_{jk}, a^q S', S_{-1}^{**}) = (1, \bar{m})$. But then the relation $u_1^{G'}(h_{ij}, S'^j(h_{ij})S', S_{-1}^{**}, v^j) < u_1^{G'}(h_{ij}, a^q S', S_{-1}^{**}, v^j)$ implied by equation A.3 contradicts $S'^j(h_{ij})$ being the uniquely justifiable action for j at I_{p1} . $\square$

⁴⁷If the game is also a personal clock auction, then at both I_1 and I_2 , there must exist one or more quitting actions. Further, each quitting action, and more generally any action leading to an item outcome (for some contingency), must always result in the outcome $(1, \bar{m})$ for some fixed $\bar{m}$.

⁴⁸Using G' weakly coarsens 1's information partition.

APPENDIX B. ADDITIONAL RESULTS (FOR ONLINE PUBLICATION ONLY)

TABLE B.1. Summary of Valuations with Censored Data

Treatment	Sample	N	Rational Valuations	Valuations >\$1	Valuations <\$0.95	Multiple Switch
OSP	Full	193	31.1%	57.5%	11.4%	-
	Additional	300	28.7%	66.0%	5.3%	-
No-Exit-OSP	Full	174	52.3%	44.3%	3.5%	-
	Additional	301	52.9%	45.2%	2.0%	-
UJS	Full	206	43.7%	10.7%	1.9%	43.7%
	Additional	300	39.7%	20.7%	1.0%	38.7%
UJ-OSP	Full	186	50%	50%	-	-
	Additional	301	44.2%	55.8%	-	-
No-Exit-UJ-OSP	Full	192	60.4%	39.6%	-	-
	Additional	301	60.8%	39.2%	-	-

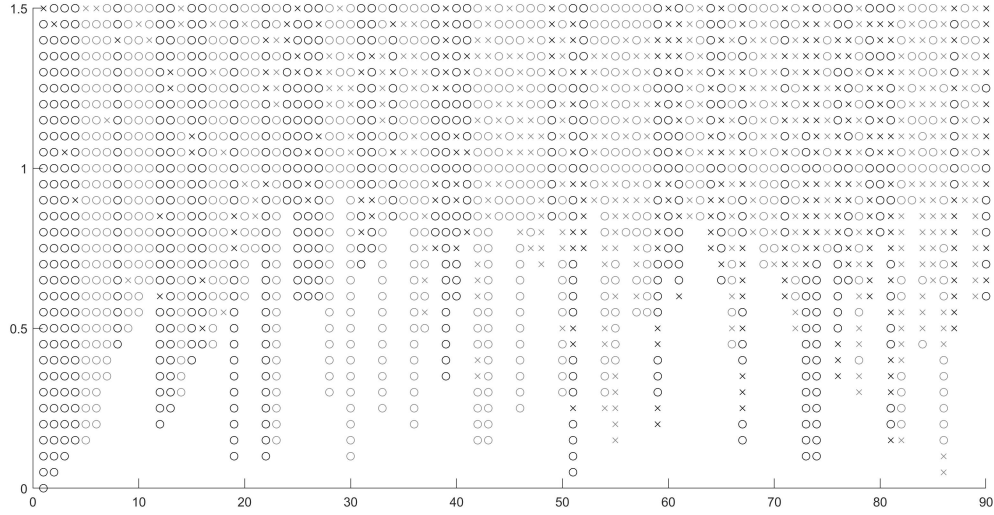
Notes: Fraction of subjects with rational valuations and those above and below \$1.00 for the Full sample used in the main analysis, and for a sample (Additional) which includes valuations derived from the incentivized question used to elicit a subject's valuation when they are censored before ever choosing the item.

TABLE B.2. Valuations in No-Exit-UJ-OSP

Treatment	Sample	N	Rational Valuations	Valuations >\$1	Valuations <\$0.95	Multiple Switch
No-Exit-UJ-OSP	Full	192	60.4%	39.6%	-	-
	Attentive	74	79.7%	20.3%	-	-

Notes: Fraction of subjects with rational valuations and those above and below \$0.95 for the No-Exit-UJ-OSP treatment. We report the results separately for the subsamples of subjects that answered all quiz questions correctly (Attentive). The UJ-OSP-D mechanism in the No-Exit-UJ-OSP treatment is a diagnostic mechanism that prevents valuations below \$1.00.

FIGURE B.1. Multiple-Switching Behavior



Notes: Multiple-switching behavior. Each circle indicates a rational choice and each cross indicates an irrational choice. At \$1.00, we treat all choices as rational (due to subjects being indifferent). Each column is the choice behavior of a single subject. Subjects are sorted from those that make the most rational choices (on the left) to the least (on the right). The darker columns indicate subjects that answered all of the quiz questions correctly while those in gray answered one or more questions incorrectly.

As with the comparison between OSP and UJ-OSP, we observe more early exits at high valuations in No-Exit-OSP relative to No-Exit-UJ-OSP. However, the difference is not statistically significant in either the Full Sample ($p = 0.212$) nor the Attentive sample ($p = 0.300$). We hypothesize that the differences between No-Exit-OSP and No-Exit-UJ-OSP were smaller because, by forcing subjects to remain in the experiment, we nudged them to engage more. As discussed in Section 5.3, we provided evidence of this conjecture through the fact that we observe less late exists in the No-Exit-OSP treatment relative to the OSP treatment.

Also consistent with this conjecture, high valuations drop by 13.2% from OSP to No-Exit-OSP in the Full sample and 18% in the Attentive sample, both of which are significant by two-sided Fisher exact tests ($p < 0.040$). In contrast, high valuations drop by 10.4% from UJ-OSP to No-Exit-UJ-OSP in the Full sample and by only 1.9% in the Attentive subsample. The difference is significant in the Full sample ($p = 0.049$), but not in the Attentive sample ($p = 0.820$). The fact that we see larger differences between OSP and No-Exit-OSP than UJ-OSP and No-Exit-UJ-OSP suggests that forcing subjects not to exit encourages introspective learning that goes beyond reducing opportunity costs. In the UJ-OSP mechanism, such learning may play less of a role given that one does not have to reason through future contingencies. As such, the comparison between OSP and UJ-OSP gives us greater power to detect failures of rationality in the OSP mechanism.

APPENDIX C. ADDITIONAL THEORETICAL DETAILS AND EXAMPLES (FOR ONLINE
PUBLICATION ONLY)

Example 9. (Posted price mechanism) Suppose there are M players, with each player having two types $\{v_h, v_l\}$. The auctioneer approaches every player sequentially in a random order, and offers them a take-it-or-leave-it offer to sell the item at a randomly drawn price (say $p^* \in (v_l, v_h)$).⁴⁹ The game ends when a player agrees to sell at the offered price. This resembles a (sequential) posted-price mechanism which has over time become hugely popular among online sellers (see Einav et al., 2018) and has been shown to be SOSp (see Theorem 8 from Pycia and Troyan (2023)). This game is also a UJS mechanism, with a simplified game identical to the original game for every participating player. Every contingency conceivable to a player is payoff-relevant, and both types of the player has a uniquely justifiable action to sell the item if $v < p^*$.

Explanation for **Example 4** (Binary Public Good): To see why the VCG mechanism is not a UJS mechanism, say for P1 under $n = 2$, suppose $v_1 = 4$ and $a_2 = v_2 = 3$, $C = 5.5$. The total reported value without P1 is $3 < C$, so without her, the project would not be built, yielding P1 a payoff of 0. With her truthful report of $a_1 = v_1 = 4$, the total is $7 \geq C$, so the project is built and her net utility is $v_1 - (C - a_2) = 4 - (5.5 - 3) = 1.5$. However, in this contingency, any report $a_1 \geq 4$ would also ensure the project is built and would yield the same payoff 1.5 to P1. Since P1’s information partition is already as coarse as possible (a single report), there is no simplified game G' that makes truthful reporting the uniquely justifiable action. Hence, the public project VCG mechanism is not a UJS mechanism.

To see why the price-list version is a UJS mechanism, fix player i and a node of her list at price $q \in \mathcal{Q}$. Here, the action is payoff-relevant *only* in contingencies where “the required price equals” q . If $v_i > q$ and $\hat{q}_i = q$, then *Yes* yields payoff $v_i - q > 0$, while *No* yields 0. Hence *Yes* strictly dominates and is the uniquely justifiable action at q . If $v_i < q$ and $\hat{q}_i = q$, then *Yes* yields $v_i - q < 0$, while *No* yields 0. Hence *No* is the uniquely justifiable action at q . Because at every node q the uniquely justifiable action (under its only payoff-relevant contingency) is *Yes* at $q \iff q < v_i$, the player’s uniquely justifiable strategy on the whole list is a threshold rule that is also a dominant strategy in the original mechanism. Therefore, this is a UJS mechanism. Furthermore, the dominant strategy equilibrium implements the VCG allocation and payments.

Non-triviality condition: One can rule out SOSp mechanisms that implement the utilitarian efficient rule, by assuming the following *non-triviality* condition: there exists some player i

⁴⁹The price can also be redrawn for each new player, but for simplicity, we focus on the same price for now.

with three types $v_i^1, v_i^2, v_i^3 \in \mathcal{M}_i$ and there exist two type-profiles of other players v_{-i}, v'_{-i} , such that the tuple

$$\left(\mathbf{1} \left\{ v_i^k + \sum_{j \neq i} v_j \geq C \right\}, \mathbf{1} \left\{ v_i^k + \sum_{j \neq i} v'_j \geq C \right\} \right)$$

takes different values for $k = 1, 2, 3$. This ensures that at least three types of some player matter for the allocation rule, ruling out trivial cases where only one cutoff over types determines the outcome.⁵⁰

TTC algorithm for Example 5: For each student, at each step, we point from the student to the acceptable school that is the most preferred by that student among the schools that are still present. If there is no such school, she points at herself. For each school, we point from the school to the student who has the highest priority for the school among the acceptable students who are still present. Every student in a cycle is assigned to the school she is pointing to or to herself and is removed. The remaining capacity of each school in the cycle is reduced by one and if it reaches zero, the school is removed. The algorithm stops when all students or all schools have been assigned.

Contingency: $s_1 : A \succ C \succ B$, $s_3 : C \succ A \succ B$.

Priorities: $\pi_A : s_1 \succ s_2 \succ s_3$, $\pi_B : s_1 \succ s_3 \succ s_2$, $\pi_C : s_1 \succ s_2 \succ s_3$.

Capacities = 1.

Each round: students point to top remaining school; schools point to top-priority remaining student; clear all cycles.

Case1 : s_2 reports $B \succ A \succ C$.

Step 1: $s_1 \rightarrow A$, $s_2 \rightarrow B$, $s_3 \rightarrow C$; $A \rightarrow s_1$, $B \rightarrow s_1$, $C \rightarrow s_1$. Cycle $(s_1, A) \Rightarrow$ assign $s_1 - A$.

Step 2: $s_2 \rightarrow B$, $s_3 \rightarrow C$; $B \rightarrow s_3$, $C \rightarrow s_2$. Cycle $(s_2, B, s_3, C) \Rightarrow$ assign $s_2 - B$, $s_3 - C$.

Final: $s_1 - A$, $s_2 - B$, $s_3 - C$.

Case2 : s_2 reports $A \succ B \succ C$.

Step 1: $s_1 \rightarrow A$, $s_2 \rightarrow A$, $s_3 \rightarrow C$; $A \rightarrow s_1$, $B \rightarrow s_1$, $C \rightarrow s_1$. Cycle $(s_1, A) \Rightarrow$ assign $s_1 - A$.

Step 2: $s_2 \rightarrow B$, $s_3 \rightarrow C$; $B \rightarrow s_3$, $C \rightarrow s_2$. Cycle $(s_2, B, s_3, C) \Rightarrow$ assign $s_2 - B$, $s_3 - C$.

Final: $s_1 - A$, $s_2 - B$, $s_3 - C$.

Case3 : s_2 reports $C \succ B \succ A$.

Step 1: $s_1 \rightarrow A$, $s_2 \rightarrow C$, $s_3 \rightarrow C$; $A \rightarrow s_1$, $B \rightarrow s_1$, $C \rightarrow s_1$. Cycle $(s_1, A) \Rightarrow$ assign $s_1 - A$.

Step 2: $s_2 \rightarrow C$, $s_3 \rightarrow C$; $B \rightarrow s_3$, $C \rightarrow s_2$. Cycle $(s_2, C) \Rightarrow$ assign $s_2 - C$.

Step 3: Remaining s_3, B : cycle $(s_3, B) \Rightarrow$ assign $s_3 - B$.

⁵⁰To see why the presence of three types does not by itself rule out an efficient SOSP mechanism, take a game with n players, each with potential valuations $v_i \in \{0, 1, 10\}$ and threshold $C = 10n$. A posted-price-style mechanism that asks every player i whether they are willing to pay $p_i \in [2, 10]$ for the project, and implements the project at the per-individual price of p_i when all say “yes,” is both SOSP and efficient.

Final: $s_1 - A$, $s_2 - C$, $s_3 - B$.

One might wonder if every SOSP mechanism is also a UJS mechanism if the players have only two types and they only make one choice.⁵¹ We provide a counterexample below, which also shows the diversity of UJS mechanisms.

Example 10. (UJS vs SOSP) Suppose a seller with quasi-linear utility has two types, one with $v^j = 10$ and another with $v^k = 20$ respectively. Consider the mechanism that asks the seller to choose between selling the item for \$15 or keeping the item. After the choice, the mechanism tosses a coin: if Heads, then the seller's choice (sell or not sell at \$15) is implemented. If Tails, the seller's choice is implemented *and* she is additionally given \$10, irrespective of her choice. The $v^k = 20$ type does not have a SOD strategy because the best case outcome from selling (\$15+\$10=\$25) exceeds the worst case outcome from not (\$20). But, the game is a UJS game, and the simplified game is identical to the original game. The mechanism does not require the seller to think through all contingencies (toss outcome): if the high type thinks the coin toss will be Heads, she should not sell, and similarly if she thinks it will be Tails. Similarly, the low type would not want to sell, irrespective of which contingency she thinks about. *Crucially, thinking about any contingency leads to finding the (right) uniquely justifiable action for both types.*

Example 11. Standard MPL as a UJS Mechanism: The standard Multiple Price List (MPL) is a UJS mechanism, even though its structure differs slightly from the UJS-E version we implement in our experiment. In UJS-E, the game ends as soon as the Nature-selected clock price is reached—this was done to align the format more closely with OSP-E. In contrast, the standard MPL requires the subject to make binary choices at every possible price, regardless of Nature's choice. Despite this difference, the standard MPL remains a UJS mechanism.

In Figure C.1, we present a shortened extensive-form version of such an MPL, assuming that the choice at $CP = 1.05$ is made before that at $CP = 0.95$. On the right pane, we show the corresponding simplified-game which offers a uniquely justifiable strategy to each type. Importantly, the logic remains unchanged even if the order of clock prices is reversed.

⁵¹Especially because we prove the mutual exclusivity of UJS and OSP mechanisms in Proposition 2 only for $N \geq 3$ types.

FIGURE C.1. MPL versus Simplified MPL

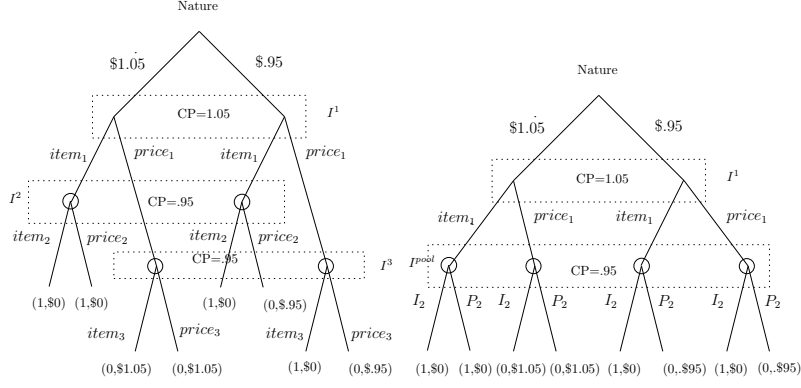


FIGURE C.2. Shortened MPL and its *simplified* game.

APPENDIX D. CALIBRATION (FOR ONLINE PUBLICATION ONLY)

We conduct a calibration exercise to assess the plausibility of rational expectations of mistakes in the OSP mechanism. Before going into the details, here is the basic intuition for why rational expectations of errors does not appear to be a plausible explanation for early exit. If subjects make mistakes uniformly, then as the probability of a mistake increases, the chance of ever reaching \$1.00 decreases because mistakes result in choosing the item, ending the task. Even if subjects only make mistakes at prices less than \$1.00, the probability of a mistake must be very large to make the expected value of continuing less than that of exiting. At a current price of \$1.50, the probability of a mistake must be greater than 0.9. At \$1.25, it must be greater than 0.75. For comparison, the average probability of a mistake in the UJS treatment among subjects that switch multiple times is only 0.27. Thus, it seems that correct beliefs can only plausibly explain choosing the item at prices just slightly greater than \$1.00, whereas we observe many subjects choosing the item at much higher prices.

Suppose a subject believes that they will make a mistake with probability γ in every future period. Denote the break-even price, $\hat{t}$: $p(\hat{t}) = 1$. The expected profit from continuing in period $t' \geq \hat{t}$ (at a price not less than \$1.00) is given by

$$\begin{aligned}
 E[\pi_t] &= \sum_{t=t'}^{\hat{t}} \frac{1}{T-t'+1} (1-\gamma)^{t-t'+1} p(t) + \sum_{t=\hat{t}+1}^T \frac{1}{T-t'+1} (1-\gamma)^{\hat{t}-t'+1} \gamma^{t-\hat{t}} p(t) + \\
 (D.1) \quad &1 - \sum_{t=t'}^{\hat{t}} \frac{1}{T-t'+1} (1-\gamma)^{t-t'+1} - \sum_{t=\hat{t}+1}^T \frac{1}{T-t'+1} (1-\gamma)^{\hat{t}-t'+1} \gamma^{t-\hat{t}}
 \end{aligned}$$

where $p(t)$ is the price in period t .

The first term is the expected profit when the subject receives a price greater than or equal to \$1.00 which occurs only when the mechanism randomly ends and the subject has not previously made a mistake (choosing the item and ending the mechanism). The second term is the expected profit when the subject receives a price less than \$1.00, which requires the mechanism to end, the subject to not have made mistakes at prices greater than \$1.00, and the subject to have made mistakes at all lower prices. The third and fourth terms are the expected profits when the subject chooses the item. They are determined by subtracting the probability of receiving the price from one.

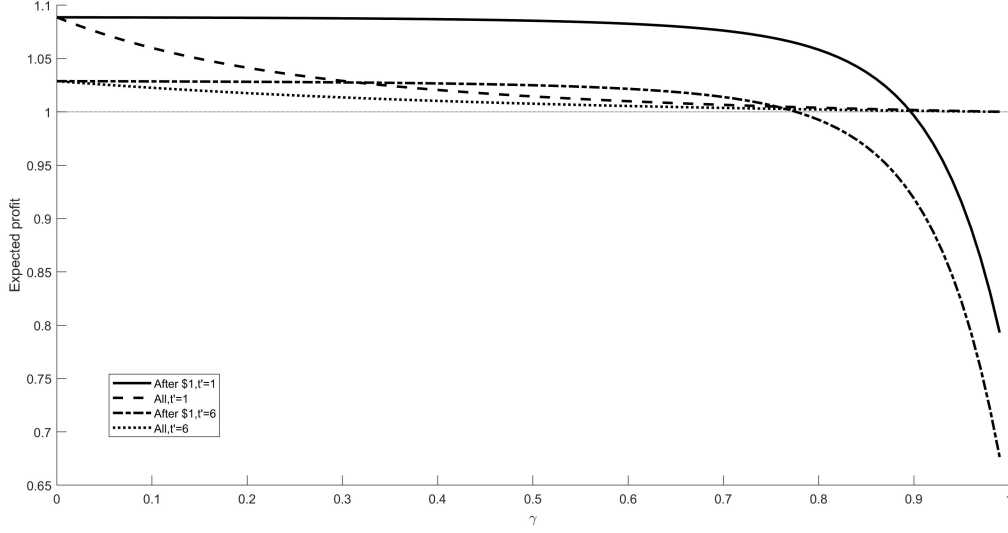
Alternatively, if a subject expects to make mistakes *only* at prices less than \$1.00, the expected profit becomes

$$\begin{aligned}
 E[\pi_t] = & \sum_{t=t'}^{\hat{t}} \frac{1}{T-t'+1} p(t) + \sum_{t=\hat{t}+1}^T \frac{1}{T-t'+1} \gamma^{t-\hat{t}} p(t) + \\
 (D.2) \quad & 1 - \sum_{t=t'}^{\hat{t}} \frac{1}{T-t'+1} - \sum_{t=\hat{t}+1}^T \frac{1}{T-t'+1} \gamma^{t-\hat{t}}
 \end{aligned}$$

after removing the probabilities of choosing the item at prices greater than or equal to \$1.00.

Figure D1 plots the expected profit for each of these two possible models as a function of the mistake probability, γ , for $t' = 1$ (current price is \$1.50) and $t' = 6$ (current price is \$1.25). ‘All’ is the first model in which mistakes are allowed at all prices. ‘After \$1’ is the second model in which mistakes are only allowed at prices less than (after) \$1.00.

Figure D1: Expected Profit with Rational Expectations



Notes: ‘All’ is for the model in which all mistakes are allowed (at any price). ‘After \$1’ is the model in which only mistakes after \$1.00 are allowed. γ is the probability of a mistake. $t' = 1$ corresponds a current price of \$1.50 and $t' = 6$ corresponds to a current price of \$1.25.

It is clear from Figure D1, that in the model in which mistakes can be made at any price, the expected profit from continuing is never less than the expected profit of choosing the item (\$1.00). Intuitively, as the probability of a mistake increases so that it is possible to get less than \$1.00, the chance of ever reaching a price of \$1.00 decreases because the subject is more likely to mistakenly choose the item before this time.

In the model in which a subject only expected to make mistakes at prices less than \$1.00, expected profits of less than \$1.00 are possible. However, they require subjects to believe they will make mistakes with very high probability. To chose the item at \$1.50, the probability of a mistake must be at least 0.9 which seems much too large to be plausible. To chose the item at \$1.25, the probability of a mistake can be lower (at least 0.75), but this still seems much too large for rational expectations. As a point of comparison, in the UJS mechanism, the average probability of a mistake among subjects that switch multiple times is only 0.27.

APPENDIX E. INSTRUCTIONS (FOR ONLINE PUBLICATION ONLY)

The following are the instructions and comprehension questions for the BDM treatment.



Instructions

In this experiment, you will complete a single task. You own an item that can be redeemed for a bonus of \$1.00. The experimenter wants to buy it. Your task is to make an offer for the item.

Your offer will be compared to an unrelated fixed offer that is equally likely to be any number between \$0.00 and \$1.50. If your offer is less than or the same as the fixed offer, then you sell the item. You had the low offer, so you are the seller. But, here's the interesting part. You do not receive your offer. Instead, you receive the fixed offer, a price higher than your offer.

Example: if you offer \$0.75 and the fixed offer is \$0.80, you have the low offer. You sell the item and you receive the fixed offer of \$0.80.

If your offer is more than the fixed offer then you do not sell your item. You keep the item.

Example: if you offer \$1.25 and the fixed offer is \$1.20, you do not have the low offer. Therefore, you do not sell the item.

You **should offer the minimum amount you would be willing to accept** in exchange for the item you own. There are no advantages to strategic behavior in this task. Your best strategy is to determine your personal value for the item and offer that value.

Why is my best strategy to bid the minimum I'd be willing to accept? Let's go back to the example:

Say I decide that the minimum I'd be willing to accept for my item is \$1.25. What happens if I ask for less than \$1.25? Say I ask for only \$1.10. If the fixed amount is, say, \$1.15, then I have to sell my item. I lose out because I have to give up the item which I think is worth \$1.25, but I only get \$1.15 in exchange.

What happens if I ask for more than \$1.25? Say I ask for \$1.50. If the fixed amount is \$1.40, then I do not sell. But, had I bid \$1.25, I would have sold the item and received \$1.40 for an item that I think is worth only \$1.25. I would have done better.

When you have understood these instructions, please proceed to the next page to answer several comprehension questions.



Comprehension Questions

Please answer the following comprehension questions. Click 'Show/hide instructions' if you need to review the instructions again.

Show/hide instructions

1. Please indicate which of the following statements is true:

- ☐ the fixed offer is more likely to be \$1.50 than \$1.00
- ☐ the fixed offer is more likely to be \$1.00 than \$1.50
- ☐ the fixed offer is equally like to be \$1.00 or \$1.50

2. What will be your bonus if the fixed offer is \$1.25 and your offer is \$1.15?

- ☐ \$1.25
- ☐ the value of the item
- ☐ \$1.15

3. What will be your bonus if the fixed offer is \$1.15 and your offer is \$1.25?

- ☐ \$1.25
- ☐ the value of the item
- ☐ \$1.15

The following are the instructions and comprehension questions for the OSP treatment. They are identical for the UJ-OSP treatment except for the single additional sentence in the instructions, “You will be forced to choose Exit the Market when the price reaches \$1.00. Therefore, you cannot earn a bonus less than \$1.00.”



Instructions

In this experiment, you will complete a single main task which we describe below. In addition, we will ask you one paid question after the main task. If you answer it correctly, we will give you an additional bonus of \$0.50. This paid question is separate from the comprehension questions which follow these instructions.

INSTRUCTIONS FOR MAIN TASK:

Period: 1

Price: \$1.50

Please choose either the left or right button.

Try to Sell

Exit the Market

In this task, you have a virtual item which is worth \$1.00.

You can try to sell your item to the computer by participating in a market where you will make choices over several periods. Each period, you will click either Try to Sell or Exit the Market, and then confirm your choice.

If you choose Exit the Market in any period, the task ends. You keep the item and can no longer sell it. In this case, your bonus is \$1.00.

If you choose Try to Sell, you offer your item for sale to the computer at the current price (shown just below the current period). The computer has already randomly selected the price at which it will buy the item from you. If it buys the item from you, you receive the current price. If not, we go to the next period, the price decreases by \$0.05, and you again choose between Try to Sell and Exit the Market.

For example, in the picture above, you are in Period 1 and the current price is \$1.50. if you click on Try to Sell, you would get \$1.50 if the computer has chosen to buy the item from you at \$1.50 (and otherwise you will go to the next period). If you instead click on Exit the Market, you would get the value of the item (\$1.00) for sure.

Each price between \$1.50 and \$0.00 is equally likely to be the price at which the computer decided to buy the item from you. In the first period, the price begins at \$1.50. It decreases by \$0.05 after each period that the computer does not buy the item. The task ends when the price decreases to \$0.00 or you choose Exit the Market, whichever comes first.

When you have understood these instructions, please proceed to the next page to answer several comprehension questions.



Comprehension Questions

Please answer the following comprehension questions. Click 'Show/hide instructions' if you need to review the instructions again.

Show/hide instructions

1. Please indicate which of the following statements is true:

- ☐ the price decreases each period
- ☐ the value of the fixed item does not change
- ☐ both of the above are true

2. What will be your bonus if you choose Try to Sell when the price is \$1.25 and the computer has chosen to buy the item from you at that price?

- ☐ \$1.25
- ☐ the value of the item
- ☐ it is random

3. What will be your bonus if you choose Exit the Market when the price is \$1.25?

- ☐ \$1.25
- ☐ the value of the item
- ☐ it is random

4. What will happen if you choose Try to Sell when the price is \$1.25 and the computer does not buy the item from you at that price?

- ☐ task continues, you can Try to Sell at \$1.20 or Exit the Market in the next period
- ☐ task ends, you earn a bonus of \$1.25
- ☐ task ends, you earn the value of the item as a bonus

The following are the instructions and comprehension questions for the No-Exit-OSP treatment. They are identical for the No-Exit-UJ-OSP treatment except for the single additional sentence in the instructions, “You will be forced to choose Exit the Market when the price reaches \$1.00. Therefore, you cannot earn a bonus less than \$1.00.” And, we added a comprehension question asking whether or not this sentence is true or false.

Instructions

In this experiment, you will complete a single main task which we describe below. In addition, we will ask you one paid question after the main task. If you answer it correctly, we will give you an additional bonus of \$0.50. This paid question is separate from the comprehension questions which follow these instructions.

INSTRUCTIONS FOR MAIN TASK:

Period: 1

Price: \$1.50

Please choose either the left or right button.

Try to Sell

Exit the Market

In this task, you have a virtual item which is worth \$1.00.

You can try to sell your item to the computer by participating in a market where you will make choices over several periods. Each period, you will click either Try to Sell or Exit the Market, and then confirm your choice.

If you choose Exit the Market in any period, you keep the item and can no longer sell it. In this case, your bonus is \$1.00.

If you choose Try to Sell, you offer your item for sale to the computer at the current price (shown just below the current period). The computer has already randomly selected the price at which it will buy the item from you. If it buys the item from you, you receive the current price and the task ends. If not, we go to the next period, the price decreases by \$0.05, and you again choose between Try to Sell and Exit the Market.

The task will continue until the price reaches the price the computer choose to buy the item from you. If you chose Exit the Market before this time, you will have to choose Exit the Market again in subsequent periods until the price reaches this price.

For example, in the picture above, you are in Period 1 and the current price is \$1.50. If you click on Try to Sell, you would get \$1.50 if the computer has chosen to buy the item from you at \$1.50 (and otherwise you will go to the next period). If you instead click on Exit the Market, you would get the value of the item (\$1.00) for sure.

Each price between \$1.50 and \$0.00 is equally likely to be the price at which the computer decided to buy the item from you. In the first period, the price begins at \$1.50. It decreases by \$0.05 after each period that the computer does not buy the item. The task ends when the price decreases to the price that the computer chose to buy the item.

When you have understood these instructions, please proceed to the next page to answer several comprehension questions.



Comprehension Questions

Please answer the following comprehension questions. Click 'Show/hide instructions' if you need to review the instructions again.

Show/hide instructions

1. Please indicate which of the following statements is true:

- ☐ the price decreases each period
- ☐ the value of the fixed item does not change
- ☐ both of the above are true

2. What will be your bonus if you choose Try to Sell when the price is \$1.25 and the computer has chosen to buy the item from you at that price?

- ☐ \$1.25
- ☐ the value of the item
- ☐ it is random

3. What will be your bonus if you choose Exit the Market when the price is \$1.25?

- ☐ \$1.25
- ☐ the value of the item
- ☐ it is random

4. What will happen if you choose Try to Sell when the price is \$1.25 and the computer does not buy the item from you at that price?

- ☐ task continues, you can Try to Sell at \$1.20 or Exit the Market in the next period
- ☐ task ends, you earn a bonus of \$1.25
- ☐ task ends, you earn the value of the item as a bonus

The following are the instructions and comprehension questions for the UJS treatment.



Instructions

In this experiment, you will complete a single main task which we describe below. In addition, we will ask you one paid question after the main task. If you answer it correctly, we will give you an additional bonus of \$0.50. This paid question is separate from the comprehension questions which follow these instructions.

INSTRUCTIONS FOR MAIN TASK:

Period: 1

Price: \$1.50

Please choose either the left or right button.

Try to Sell

Do Not Sell

In this task, you have a virtual item which is worth \$1.00.

You can try to sell your item to the computer by participating in a market where you will make choices over several periods. Each period, you will click either Try to Sell or Do Not Sell, and then confirm your choice.

If you choose Try to Sell, you offer to sell your item to the computer at the current price (shown just below the period).

The computer has already randomly chosen the price at which it will try to buy the item from you. At every other price, whether you choose Try to Sell or Do Not Sell, you will go to the next period and again choose Try to Sell or Do Not Sell.

The task will continue until the current price reaches the price the computer chose to try to buy the item from you. At that price:

- If you choose Try to Sell, you receive the current price and the task ends.
- If you choose Do Not Sell you keep the item and the task ends. In this case, your bonus is \$1.00.

For example, in the picture above, you are in Period 1 and the current price is \$1.50. If the computer tries to buy from you at this price, and you click on Try to Sell, you would get \$1.50. If you instead click on Do Not Sell, you would get the value of the item (\$1.00). If the computer does not try to buy from you, you will go to the next period.

Each price between \$1.50 and \$0.00 is equally likely to be the price at which the computer decided to try to buy the item from you. In the first period, the price begins at \$1.50. It decreases by \$0.05 after each period. The task ends when the price decreases to the price that the computer chose to try to buy the item.

When you have understood these instructions, please proceed to the next page to answer several comprehension questions.



Comprehension Questions

Please answer the following comprehension questions. Click 'Show/hide instructions' if you need to review the instructions again.

Show/hide instructions

1. Please indicate which of the following statements is true:

- ☐ the price decreases each period
- ☐ the value of the fixed item does not change
- ☐ both of the above are true

2. What will be your bonus if you choose Try to Sell when the price is \$1.25 and the computer has chosen to try to buy the item from you at that price?

- ☐ \$1.25
- ☐ the value of the item
- ☐ it is random

3. What will be your bonus if you choose Do Not Sell when the price is \$1.25 and the computer has chosen to try to buy the item from you at that price?

- ☐ \$1.25
- ☐ the value of the item
- ☐ it is random

4. What will happen if you choose Try to Sell when the price is \$1.25 and the computer does not try to buy the item from you at that price?

- ☐ task continues, you can Try to Sell at \$1.20 or Do Not Sell in the next period
- ☐ task ends, you earn a bonus of \$1.25
- ☐ task ends, you earn the value of the item as a bonus

The following are the instructions and comprehension questions for the OSP-L treatment.



Instructions

In this experiment, you will complete 3 repetitions of a main task. The instructions for the task are given below. After you complete all 3 tasks, one will be chosen randomly (with equal chance of each) and you will be paid a bonus equal to your payoff on that task. In addition, we will ask you one paid question after the first repetition of the main task. If you answer it correctly, we will give you an additional bonus of \$0.50. This paid question is separate from the comprehension questions which follow these instructions.

INSTRUCTIONS FOR MAIN TASK:

Period: 1

Price: \$1.50

Please choose either the left or right button.

Try to Sell

Exit the Market

In this task, you have a virtual item which is a lottery ticket that pays \$1.20 with a 50% chance and \$1.00 otherwise. To help understand its value, you can imagine a box with balls numbered 1 to 100 in it. The computer will randomly choose one ball. If the ball has a number less than or equal to 50 you will receive \$1.20 and otherwise you will receive \$1.00.

You can try to sell your item to the computer by participating in a market where you will make choices over several periods. Each period, you will click either Try to Sell or Exit the Market, and then confirm your choice.

If you choose Exit the Market in any period, the task ends. You keep the item and can no longer sell it. In this case, your payoff is determined by the item.

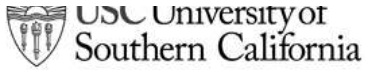
If you choose Try to Sell, you offer your item for sale to the computer at the current price (shown just below the current period). The price begins at \$1.50 in the first period and decreases by \$0.05 after each period that the computer does not buy the item. The computer has already randomly selected the price at which it will buy the item from you (each price between \$1.50 and \$0.00 is equally likely).

If the computer buys the item from you, your payoff is the current price. If not, we go to the next period, the price decreases by \$0.05, and you again choose between Try to Sell and Exit the Market.

For example, in the picture above, you are in Period 1 and the current price is \$1.50. If you click on Try to Sell, you would get \$1.50 if the computer has chosen to buy the item from you at \$1.50 (and otherwise you will go to the next period). If you instead click on Exit the Market, you would get the payoff determined by the virtual item.

You will be forced to choose Exit the Market when the price reaches \$0.05. Therefore, you cannot earn a payoff for the task less than \$0.05.

When you have understood these instructions, please proceed to the next page to answer several comprehension questions.



Comprehension Questions

Please answer the following comprehension questions. Click 'Show/hide instructions' if you need to review the instructions again.

Show/hide instructions

1. Please indicate which of the following statements is true:

- ☐ the price decreases each period
- ☐ the virtual item does not change
- ☐ both of the above are true

2. What will be your payoff if you choose Try to Sell when the price is \$1.25 and the computer has chosen to buy the item from you at that price?

- ☐ \$1.25
- ☐ it is determined by the virtual item
- ☐ \$0

3. What will be your payoff if you choose Exit the Market when the price is \$1.25?

- ☐ \$1.25
- ☐ it is determined by the virtual item
- ☐ \$0

4. What will happen if you choose Try to Sell when the price is \$1.25 and the computer does not buy the item from you at that price?

- ☐ task continues, you can Try to Sell at \$1.20 or Exit the Market in the next period
- ☐ task ends, you earn a payoff of \$1.25
- ☐ task ends, you earn a payoff that depends on the virtual item

5. True or False? You must choose Exit the Market when the price reaches \$0.05 meaning that you can not earn less than \$0.05 for the task.

- ☐ true